6 Pain Control*

JOHN M. NUSSTEIN, MELISSA DRUM, and ASMA A. KHAN

CHAPTER OUTLINE

LOCAL ANESTHESIA
Mechanisms of Action for Anesthetics
Clinically Available Local Anesthetics
Selection of a Local Anesthetic: Possible Adverse Effects and Medical History
Possible Adverse Effects
- Cardiovascular Reactions
- Systemic Effects
- Methemoglobinemia
- Peripheral Nerve Paresthesia
- Allergic Reactions

Effects of Systemic Diseases or Conditions on Local Anesthetics
Clinical Factors
Inferior Alveolar Nerve Block
2% Lidocaine and 1:100,000 Epinephrine
Anesthetic Success, Duration, and Difficulty
Alternative Anesthetic Solutions for the Inferior Alveolar Nerve Block
Solutions Without Epinephrine
Articaine With 1:100,000 or 1:200,000 Epinephrine
Long-Acting Anesthetics
Buffered Lidocaine
Diphenhydramine as a Local Anesthetic Agent
Alternative Injection Sites
Gow-Gates and Vazirani-Akinosi Techniques
Articaine Infiltrations
Volume and Concentration
Factors in Failure of the Inferior Alveolar Nerve Block
Speed of Injection and Success
Supplemental Anesthesia for Vital Pulps in the Mandible
Indications
Infiltrations
- Supplemental Mandibular Buccal, Lingual, or Buccal Plus Lingual Infiltrations of Articaine

Intraosseous Anesthesia
- Success and Duration
- Success in Symptomatic Irreversible Pulpitis
- Systemic Effects

Intraligament Anesthesia
- Success, Onset, and Duration
- Success in Symptomatic Irreversible Pulpitis
- Mechanism of Action
- Back Pressure and Amount of Solution Delivered
- Anesthetic Solutions
- Postoperative Discomfort
- Systemic Effects
- Other Factors
- Safety of the Periodontium and Pulp
- Intraseptal Injection
- Intrapulpal Injection

Maxillary Anesthesia
Alternative Anesthetic Solutions for Maxillary Infiltrations
Plain Solutions: 3% Mepivacaine (Carbocaine, Polocaine, Scandonest) and 4% Prilocaine (Citanest Plain)
4% Prilocaine With 1:200,000 Epinephrine (Citanest Forte), 2% Mepivacaine With 1:20,000 Levonordefrin (Carbocaine With Neo-Cobefrin), and 4% Articaine With 1:100,000 Epinephrine (Septocaine, Articadent, Zorcaine)
0.5% Bupivacaine With Epinephrine (Marcaine)
Extending the Duration of Pulpal Anesthesia for Maxillary Teeth
Alternative Maxillary Injection Techniques
Management of Anesthesia in Endodontic Cases
Summary and Future Directions for Effective Anesthesia
ANALGESICS AND THERAPEUTIC RECOMMENDATIONS

*Any discussion of local anesthesia over the last 40 years would inevitably lead to conversations surrounding the work of Dr. Al Reader. Although this chapter no longer bears his name, it is based on his tireless efforts, astute research mind, and never-ending quest to provide positive patient experiences in dentistry through clinical research. The authors would like to thank Dr. Reader for all of his contributions to local anesthesia and for being the guiding force that has inspired countless others to continue in the search for better local anesthesia and pain control.

Nonnarcotic Analgesics
Limitations and Drug Interactions
Acetaminophen
Opioid Analgesics
Corticosteroids
Intracanal Administration
Systemic Administration
Antibiotics
Pain Management Strategies
Pretreatment
Long-Acting Local Anesthetics
Flexible Plan
Future Directions
Summary

LOCAL ANESTHESIA

Effective local anesthesia is the foundation of pain control in dentistry and especially endodontics. Regardless of the clinician's skills, both treatment and patient management are difficult or impossible to deliver without effective pain control. This chapter reviews the pharmacology of local anesthetics and the relative advantages and limitations of various anesthetics and routes of administration. Other chapters in this book provide complementary information on the use of local anesthetics in diagnosis (see Chapter 1) and the treatment of emergency patients (see Chapter 19). The authors assume that the reader is familiar with various anesthetic injection techniques; several excellent texts are available for review regarding this topic.[165,213,277]

Mechanisms of Action for Anesthetics

Most dental pharmacology courses teach that local anesthetics block sodium channels by partitioning into two types: the uncharged basic form of the molecule (RN), which crosses cell membranes, and the charged acid form of the molecule (RNH^+), which binds to the inner pore of the sodium channel. As a first approximation, this model is reasonably accurate. However, molecular research has demonstrated the existence of at least nine subtypes of voltage-gated sodium channels (VGSCs) that differ in their expression pattern, biophysical properties, and roles in mediating peripheral pain (Table 6.1). These channels have a clear clinical relevance.[30,146,198] Indeed, several groups of patients have been described with genetic mutations to a VGSC, with significant reported effects on pain sensitivity.

The broad class of VGSCs can be divided into channels that are blocked by a toxin (tetrodotoxin [TTX]) and those that are resistant to the toxin (TTX-R). Most TTX-R channels are found primarily on nociceptors (e.g., Na_v 1.8 and Na_v 1.9).[349] These channels also are relatively resistant to local anesthetics and are sensitized by prostaglandins (PGs).[127] As explained later in this chapter, the presence of TTX-R sodium channels may explain why local anesthetics are less effective when they are administered to patients with odontalgia. Many of the adverse effects of local anesthetics are attributed to their ability to block other VGSCs expressed in the central nervous system (CNS) or heart (see Table 6.1).

VGSCs consist of an alpha and a beta subunit. The alpha subunit serves as a voltage sensor, leading to channel activation and sodium ion passage when the channel detects an electrical field. The biologic basis for an electric pulp tester (EPT), therefore, is the generation of a small electrical field across the dental pulp that can activate VGSCs.[146] Interestingly, sensitization of TTX-R channels by PGs lowers the activation threshold and increases the number of sodium ions that flow through the channel.[129] Put another way, an inflammation-induced elevation in PG levels sensitizes TTX-R channels, leading to greater activation with weaker stimuli. This may explain the increased responsiveness to electrical pulp testing seen in patients with irreversible pulpitis.

Local anesthetics have other mechanisms that may contribute to their pharmacology for treating odontogenic pain. For example, local anesthetics modulate certain G protein–coupled receptors (GPCRs). The GPCRs are a major class of cell membrane receptors, and many classes of dental drugs (e.g., opioids, catecholamines) and endogenous mediators produce their effects by activating specific GPCRs and their related second messenger pathways. Studies suggest that local anesthetics inhibit the G-alpha-q (Gaq) class of GPCRs, which includes receptors activated by inflammatory mediators such as bradykinin.[157] A local anesthetic may therefore block the actions of a major hyperalgesic agent.

Other studies have indicated that local anesthetics potentiate the actions of the G-alpha-i (Gai) class of GPCRs.[22] This could have a major effect in potentiating the actions of vasoconstrictors, including the newly recognized analgesic role that vasoconstrictors play in inhibiting pulpal nociceptors.[33,145] Prolonged alteration of GPCR function

Table 6.1 Voltage-Gated Sodium Channels and Pain

Channel Subtype	Tissue Expression	Tetrodotoxin Sensitive	Peripheral Role in Pain
Na_v 1.1	Central nervous system (CNS), sensory neurons	Yes	?
Na_v 1.2	CNS	Yes	No
Na_v 1.3	CNS	Yes	No
Na_v 1.4	Muscle	Yes	No
Na_v 1.5	Heart	Somewhat	No
Na_v 1.6	CNS, sensory neurons	Yes	?
Na_v 1.7	CNS, sensory neurons	Yes	?
Na_v 1.8	Sensory neurons	No	Yes
Na_v 1.9	Sensory neurons	No	Yes

might explain why analgesia obtained with long-acting local anesthetics persists well beyond the period of anesthesia.[72,91,237] More research is needed on this exciting aspect of local anesthetic pharmacology.

Clinically Available Local Anesthetics

The most common forms of injectable local anesthetics are in the amide class. In 2003, the American Dental Association specified a uniform color code for dental cartridges to prevent confusion among brands (Table 6.2). Local anesthetics can be divided roughly into three types: short duration (30 minutes of pulpal anesthesia), intermediate duration (60 minutes of pulpal anesthesia), and long duration (over 90 minutes of pulpal anesthesia). However, clinical anesthesia does not always follow these guidelines, as it depends on whether the local anesthetic is used as a block or for infiltration. For example, bupivacaine is classified as a long-acting agent; when it is used in an inferior alveolar nerve (IAN) block, this is true.[98] However, when it is used for infiltration for anterior teeth, it has a shorter duration of anesthetic action than 2% lidocaine with 1:100,000 epinephrine[74,134] (this is discussed in more detail later in this chapter).

Selection of a Local Anesthetic: Possible Adverse Effects and Medical History

POSSIBLE ADVERSE EFFECTS

Possible adverse reactions to local anesthetics can be divided into six major categories: cardiovascular reactions, systemic effects, methemoglobinemia, peripheral nerve paresthesia, allergic reactions to the anesthetic and/or latex, and reactions to anesthetics containing a sulfite antioxidant. These reactions range from fairly common (e.g., tachycardia after intraosseous [IO] injection of 2% lidocaine with 1:100,000 epinephrine) to extremely rare (e.g., allergic reactions to lidocaine).

Cardiovascular Reactions

Although classic research studies have reported that large dosages or intravenous injections of local anesthetics were required to produce cardiovascular effects,[163,336] it now is well recognized that even comparatively small amounts of epinephrine can induce measurable tachycardia after nerve block or IO injection.[99,129,282] Several authors have reported increases in heart rate with infiltration injections and nerve blocks using 2% lidocaine with 1:100,000 epinephrine[2,149,195,298,335]; others have reported that no significant changes in heart rate occurred or that the changes were clinically insignificant.[228,330,338] When specific information was given on dosing and heart rate increases, several studies found mean heart rate increases.[2,149,194,335] Two studies found increases on average of about 4 beats/min with approximately 20 μg of epinephrine[149,194]; three studies recorded increases of 10 to 15 beats/min with 45 to 80 μg of epinephrine[2,194,298]; and one study found increases of approximately 21 beats/min using 144 μg.[335] Increasing the amount of epinephrine in an infiltration or block injection, therefore, increases the likelihood of an elevated heart rate.

Tachycardia after injection is primarily a pharmacologic effect. The cardiovascular effects are the result of alpha-adrenoceptor stimulation by systemic distribution of the vasoconstrictor throughout the vascular compartment. The patient may also report heart palpitations associated with anxiety or fear and may experience transient tachycardia and changes in blood pressure. Large doses or inadvertent intravenous injection may lead to lidocaine toxicity and CNS depression.[100,244] To reduce this risk, the clinician should always aspirate before making the injection, inject slowly, and use dosages within accepted guidelines. The maximal dosages for local anesthetics are listed in Table 6.2.

Table 6.2 Local Anesthetics Available in the United States*

Anesthetic	Vasoconstrictor	Dental Cartridge Color Codes†	Maximum Allowable Dose	Typical Maximum Dose
2% Lidocaine	1:100,000 epinephrine	Red	13	8
2% Lidocaine	1:50,000 epinephrine	Green	13	8
2% Lidocaine	Plain (no vasoconstrictor)	Light blue	8	8
2% Mepivacaine	1:20,000 levonordefrin	Brown	11	8
3% Mepivacaine	Plain (no vasoconstrictor)	Tan	7	5½
4% Prilocaine	1:200,000 epinephrine	Yellow	5½	5½
4% Prilocaine	Plain (no vasoconstrictor)	Black	5½	5½
0.5% Bupivacaine	1:200,00 epinephrine	Blue	10	10
4% Articaine	1:100,000 epinephrine	Gold	7	7
4% Articaine	1:200,000 epinephrine	Silver	7	7

*This table provides the maximum dosage in two formats. The maximum allowable dose is generally approached only with complex oral and maxillofacial surgical procedures. The typical maximum dose is the usual outer envelope of drug dosage for most endodontic, surgical, and restorative dental procedures. Both columns show the number of cartridges that would be required for an adult weighing 67.5 kg (150 pounds).
†Uniform dental cartridge color codes were mandated by the American Dental Association in June 2003.

Systemic Effects

Acute toxicity from an overdose of a local anesthetic is often the result of inadvertent intravenous administration or of a cumulative large dose (e.g., repeated injections). As shown in Table 6.1, VGSCs are found in the CNS and the myocardium, the two major sites of anesthetic-induced toxicity. Although systemic effects from a local anesthetic are rare, they can include an initial excitatory phase (e.g., muscle twitching, tremors, grand mal convulsions) and a subsequent depressive phase (e.g., sedation, hypotension, and respiratory arrest).[78,100] It should be noted that symptomatic management (possibly including cardiopulmonary resuscitation [CPR], airway support, and supplemental oxygen) is the primary response to this adverse event.[191,193] An acute hypotensive crisis with respiratory failure has also been interpreted as the result of hypersensitivity to local anesthetics[55]; these patients should be evaluated with allergy testing. To reduce the risk of systemic effects from anesthetics, the clinician must always aspirate before giving the injection and must use dosages within accepted guidelines (see Table 6.2). Finder and Moore[100] proposed a "rule of 25" as a simple means of remembering maximal local anesthetic dosages: with currently formulated local anesthetic cartridges, it is generally safe to use one cartridge of local anesthetic for every 25 pounds of patient weight (e.g., six cartridges for a patient weighing 150 pounds [67.5 kg]).

Methemoglobinemia

The metabolism of certain local anesthetics (e.g., prilocaine, benzocaine, articaine, and to a lesser extent lidocaine) can produce a metabolite that causes methemoglobinemia; this effect often occurs several hours after injection of the local anesthetic.[214,354] Typical signs and symptoms include cyanosis, dyspnea, emesis, and headache. In a study on benzocaine-induced methemoglobinemia, 67% of reported adverse effects with benzocaine were associated with methemoglobinemia; of these events, 93% occurred with spray formulations and only one case involved the gel formulation.[239] To reduce the risk of methemoglobinemia, clinicians should take care to refrain from giving excessive dosages of local anesthetics.

Peripheral Nerve Paresthesia

Postinjection paresthesia is a rare adverse effect of local anesthetics.[136,214,365] The incidence of paresthesia (which involved the lip and/or tongue) associated with articaine and prilocaine was higher than that found with either lidocaine or mepivacaine.[114,121,136] Another study evaluated patients referred with a diagnosis of damage to the inferior alveolar and/or lingual nerve that could only have resulted from an IAN block; it was found that 35% were caused by lidocaine and 30% by articaine.[272] The conclusion was that there was not a disproportionate involvement from articaine, although this interpretation does not account for large differences in clinical use of the two solutions. This same author published a paper on IAN block involvement with the following proportions: 25% lidocaine, 33% articaine, and 34% prilocaine.[271] Another investigative group also noted more occurrences of paresthesia with articaine and prilocaine.[269] Paresthesia can be caused by any anesthetic agent. Since paresthesia is noted in the literature to be higher with articaine and because articaine does not have higher pulpal anesthetic success rates when given as an IAN block, articaine use may be best reserved for infiltrations. With any alteration in nerve sensation, documentation of the area of altered sensation (patient reported and clinically tested), the type of altered sensation (e.g., anesthesia, paresthesia, dysesthesia), and regular follow-up are important.[371]

Allergic Reactions

The amide local anesthetics appear to have little immunogenicity and therefore have an extremely low rate of true allergic reactions,[20,244,303,292] but case reports of hypersensitivity reactions after the administration of local anesthetics have been published.[32,55,141,240,303,310,318] Referral for allergy testing (with dental anesthetic cartridges) or deep sedation may be indicated in patients with true allergic reactions (anaphylaxis, etc.).[303]

Local anesthetic formulations that contain vasoconstrictors also contain a sulfite to prevent oxidation of the agent. The reaction to sulfites in dental anesthetics may be different from the reaction caused by foods.[318] This reaction is unlikely to occur, but the use of local anesthetics without vasoconstrictors is a possible alternative in patients with severe sulfite allergies.

Effects of Systemic Diseases or Conditions on Local Anesthetics

It has been stated that vasoconstrictors should be avoided in patients with high blood pressure (higher than 180 mm Hg systolic or 120 mm Hg diastolic), cardiac dysrhythmias, unstable angina, less than 6 months since myocardial infarction or cerebrovascular accident, or severe cardiovascular disease.[213] However, these conditions are contraindications to *routine* dental treatment.

Local anesthetics may interact with a patient's medications, so a thorough review of the medical history is an absolute requirement. Potential drug-drug interactions occur primarily with the vasoconstrictors in local anesthetic formulations (Table 6.3). Judicious use of local anesthetic solutions without vasoconstrictors (e.g., 3% mepivacaine) is a reasonable alternative for those adult patients.

Many of the commonly available local anesthetics are safe for use in pregnant or lactating women.[137] The US Food and Drug Administration (FDA) has classified lidocaine and prilocaine as category B and mepivacaine, bupivacaine, and articaine as category C. The point in pregnancy (trimester) and emergent need for care need to be considered. The most important aspect of care with pregnant patients is to eliminate the source of pain by performing the indicated endodontic treatment.[137]

Clinical Factors

Traditional methods of confirming anesthesia usually involve questioning the patient, probing the soft tissue, and/or starting treatment and waiting for the patient's reaction. These approaches may not be effective for determining pulpal anesthesia.[53,154,225,340]

Table 6.3 Possible Drug Interactions With Vasoconstrictors

Drugs	Possible Adverse Effects	Recommendations
TRICYCLIC ANTIDEPRESSANTS		
Amitriptyline, doxepin	Increased cardiovascular responses	Reduce or eliminate vasoconstrictors
NONSELECTIVE BETA BLOCKERS		
Nadolol, propranolol	Hypertension, bradycardia	Reduce or eliminate vasoconstrictors
RECREATIONAL DRUGS		
Cocaine	Hypertension, myocardial infarction, dysrhythmias	Instruct patient to abstain from drug use for 48 h before procedure; do not use vasoconstrictors
COMT INHIBITORS		
Entacapone, tolcapone	Increased cardiovascular responses	Reduce or eliminate vasoconstrictors
ANTIADRENERGIC DRUGS		
Guanadrel, guanethidine	Increased cardiovascular responses	Reduce or eliminate vasoconstrictors
NONSELECTIVE ALPHA-ADRENERGIC BLOCKERS		
Chlorpromazine, clozapine, haloperidol	Increased cardiovascular responses	Reduce or eliminate vasoconstrictors
DIGITALIS		
Digoxin	Dysrhythmias (especially with large dosage of vasoconstrictor)	Reduce or eliminate vasoconstrictor
HORMONE		
Levothyroxine	Dysrhythmias (especially with large dosage of vasoconstrictor)	*Euthyroid:* No precaution *Hyperthyroid:* Reduce or eliminate vasoconstrictors
MONOAMINE OXIDASE INHIBITORS		
Furazolidone, linezolid, selegiline, tranylcypromine	No interaction	None

COMT, Catecholamine *O*-methyl transferase.
Modified from Naftalin L, Yagiela JA: Vasoconstrictors: indications and precautions, *Dent Clin North Am* 46:733, 2002.

Anesthesia in asymptomatic vital teeth can be measured more objectively by applying a cold refrigerant (Fig. 6.1) or by using an EPT, as described in Chapter 1 (Fig. 6.2). Application of cold or the EPT can be used to test the tooth under treatment for pulpal anesthesia before a clinical procedure is started.[49,88,170,198,210]

In symptomatic vital teeth (after administration of a local anesthetic), testing with a cold refrigerant or an EPT can also be used to evaluate pulpal anesthesia before starting a treatment.[62,88,253,357] If there is a positive response to testing, supplemental anesthesia will be necessary. However, in these particular patients, a negative response does not guarantee pulpal anesthesia,[92,259,349] and supplemental anesthesia may still be necessary. If the pulp chamber contains necrotic tissue and the canals have vital tissue, no objective test can predict the level of pulpal anesthesia.

Why patients in pain have difficulty with anesthesia is a complicated question. If there was an easy answer, we would have an easy way to provide predictable profound anesthesia for every patient every time. Theories related to pH, hyperalgesia, altered resting potentials and lowered excitability thresholds, increased sodium channel expression, TTX-R (resistant) sodium channels, and lowered pain thresholds have been suggested as potentially compromising the efficacy of local anesthetics.[146,44,343,230,293,320,347,349] This is different from the central core theory, which has been offered as an explanation for failure with the IAN block in asymptomatic vital teeth.[77,325] This theory relates to the inner portion of the nerve bundle supplying the anterior teeth and the outer portion of the nerve bundle supplying the posterior teeth (Fig. 6.3), explaining part of the higher failure rates in anterior teeth.



Fig. 6.1 A cold refrigerant may be used to test for pulpal anesthesia before the start of a clinical procedure. (Courtesy Coltène/Whaledent Inc., Cuyahoga Falls, OH.)



Fig. 6.2 An electric pulp tester also may be used to test for pulpal anesthesia before a clinical procedure is started. (Courtesy SybronEndo, Corporation Orange, CA.)

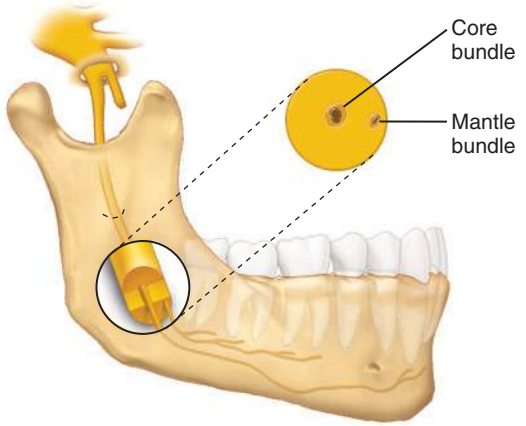


Fig. 6.3 The central core theory. The axons in the mantle bundle supply the molar teeth, and those in the core bundle supply the anterior teeth. The extraneural local anesthetic solution diffuses from the mantle to the core. (Modified and redrawn from De Jong RH: *Local anesthetics,* St Louis, 1994, Mosby.)

Inferior Alveolar Nerve Block

2% LIDOCAINE AND 1:100,000 EPINEPHRINE

Because failure occurs most often with the IAN block,[180] factors that modify mandibular anesthesia must be carefully reviewed. The technique for administering an IAN block can be reviewed in available textbooks.[165,213] The following discussion reviews the expected outcomes after administration of a conventional IAN block to asymptomatic patients using 1.8 mL of 2% lidocaine with 1:100,000 epinephrine (Xylocaine, Lignospan, Octocaine). Although anesthesia requirements vary among dental procedures, the following discussion concentrates on *pulpal* anesthesia in asymptomatic patients.

ANESTHETIC SUCCESS, DURATION, AND DIFFICULTY

Anesthetic success for nerve blocks has been defined in the majority of studies as the percentage of subjects who achieve two consecutive nonresponsive readings on electrical pulp testing within 15 minutes and continuously sustain this lack of responsiveness for 60 minutes. The clinical condition would be pulpal anesthesia within 15 minutes that lasts for 1 hour. Using this criterion, the percentage of cases in which anesthesia was obtained after IAN block injections ranged from 10% (central incisor) to 65% (second molar).[53,140,154,225,252,277,340] In all of these studies, patients reported profound lip numbness, meaning that these are not "missed" blocks. Profound lip numbness *does not* predict pulpal anesthesia. However, lack of soft-tissue anesthesia is a useful indicator that the block injection was not administered accurately for that patient. This will occur approximately 4% to 6% of the time in asymptomatic patients.[106] The authors also found that administration of a two-cartridge volume was significantly better for avoiding a missed block than a one-cartridge volume in asymptomatic subjects. It is important to note, however, that a two-cartridge volume does not provide a higher success rate for pulpal anesthesia.[106,341,366] Once pulpal anesthesia does occur, it usually persists for approximately 2½ hours.[98] Fig. 6.4 depicts the time course for complete pulpal anesthesia for an asymptomatic first molar, as defined by the percentage of patients who did not respond to a stimulus (EPT) across time for 60 minutes. Most patients achieved pulpal anesthesia within 15 minutes and had a duration of anesthesia of at least 1 hour, but the success rate was not 100%.

Anesthetic failure can be defined as the percentage of subjects who never achieved two consecutive nonresponsive EPT readings at any time during a 60-minute period. Using this criterion, anesthetic failure rates range from 17% (second molar) to 58% (central incisor).[53,98,140,154,225,252,277,309,340,341]

Another presentation of mandibular anesthesia is noncontinuous anesthesia, which may be related to the action of the anesthetic solution on the nerve membrane (blocking and unblocking the sodium channels). This occurs in about 12% to 20% of patients.[53,135,140,154,225,252,277,340]

After a conventional IAN block injection, the onset of pulpal anesthesia occurs within 10 to 15 minutes in most cases (see Fig. 6.4).[53,98,140,154,225,252,277,309,340,341] Slow onset of anesthesia can occur in a percentage of subjects who are nonresponsive to the EPT after 15 minutes. In mandibular teeth, slow onset occurs in 12% to 20% of patients.

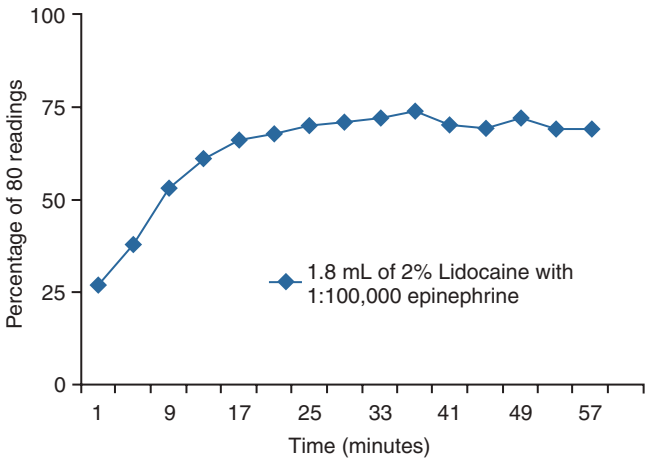


Fig. 6.4 Incidence of first mandibular molar anesthesia as determined by lack of response to electrical pulp testing at the maximum setting (percentage of 80 readings) across time for 60 minutes.

Alternative Anesthetic Solutions for the Inferior Alveolar Nerve Block

SOLUTIONS WITHOUT EPINEPHRINE

Plain solutions (mepivacaine, prilocaine) have been found to obtain pulpal anesthesia for an IAN block at rates similar to those for lidocaine with 1:100,000 epinephrine.[62] These findings support the selection of 3% mepivacaine as a local anesthetic when medical conditions or drug therapies suggest caution in the administration of solutions containing epinephrine; however, the duration of pulpal anesthesia is reduced to approximately 50 minutes.[225]

ARTICAINE WITH 1:100,000 OR 1:200,000 EPINEPHRINE

Articaine is classified as an amide. It has a thiophene ring (instead of a benzene ring like the other amide local anesthetics) and an extra ester linkage, which results in hydrolysis of articaine by plasma esterases.[219] Lidocaine and articaine have the same maximal dose of 500 mg for adult patients (recommended dose, 6.6 to 7 mg/kg), but the maximum number of cartridges is different because of the differences in drug concentration (see Table 6.2).[213]

Many clinical trials and some meta-analyses have failed to demonstrate any statistical superiority of articaine over lidocaine for IAN blocks in both symptomatic and asymptomatic patients.[17,35,57,197,273,311,319,332]

LONG-ACTING ANESTHETICS

Clinical trials with bupivacaine (Marcaine) have been conducted in patients undergoing oral surgery,[76,292] endodontic treatment,[91,98,237,262,299], and periodontic treatment.[71,207] Bupivacaine was found to have a slower onset of pulpal anesthesia than lidocaine for IAN blocks.[98] Generally, bupivacaine provides prolonged analgesia and is indicated when postoperative pain is anticipated, but not all patients want lip numbness for an extended period.[292] Patients should be questioned about their preference. Although bupivacaine has a somewhat slower onset than lidocaine, its duration of pulpal anesthesia in the mandible is almost twice as long (4 hours) (Fig. 6.5).[98]

Ropivacaine (Naropin), a relatively new long-acting local anesthetic, is a structural homolog of bupivacaine.[187] It is currently unavailable in dental cartridge form. A number of studies have shown that ropivacaine has a lower potential for toxic CNS and cardiovascular effects than bupivacaine but produces equivalent pharmacologic effects.[187] Ropivacaine and levobupivacaine are being developed as potentially new local anesthetics based on their stereochemistry. Both are S-isomers and are thought to cause less toxicity than the racemic mixture of bupivacaine currently marketed.[312] A medical clinical trial has indicated that levobupivacaine showed significantly better postoperative pain control at 4 and 24 hours after infiltration injection than ropivacaine.[259,215]

Liposomal bupivacaine (Exparel, Pacira Pharmaceuticals, San Diego, CA) has been approved for infiltration use by the FDA. The formulation is a preservative-free aqueous suspension of multivesicular liposomes (DepoFoam) containing bupivacaine at a concentration of 13.3 mg/mL (expressed as anhydrous bupivacaine hydrochloride [HCl] equivalent).[258,275] A small amount (3%) is free bupivacaine, which provides immediate numbness, whereas the remaining drug is released from the liposomes over time.[275] The FDA has *not* approved Exparel for nerve blocks. In clinical endodontic studies, Exparel did not significantly reduce symptomatic irreversible pulpitis pain or pain medication usage as compared with bupivacaine in untreated cases.[41] In another endodontic clinical trial using symptomatic patients diagnosed with pulpal necrosis who were experiencing moderate to severe preoperative pain and receiving endodontic treatment, a 4.0-mL infiltration of liposomal bupivacaine did not result in a statistically significant decrease in posttreatment pain compared with an infiltration of 4.0 mL of 0.5% bupivacaine with 1:200,000 epinephrine.[126]

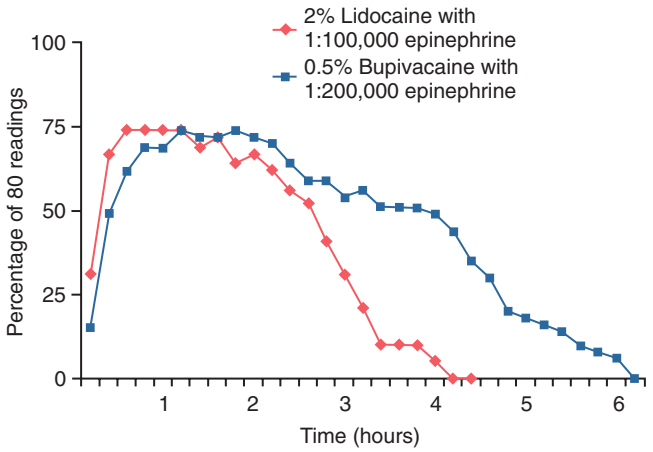


Fig. 6.5 Incidence of first mandibular molar anesthesia: comparison of 0.5% bupivacaine with 1:200,000 epinephrine to 2% lidocaine with 1:100,000 epinephrine. Results were determined by lack of response to electrical pulp testing at the maximum setting (percentage of 80 readings) across time for 6 hours. The bupivacaine solution showed a longer duration of anesthesia than the lidocaine solution.

BUFFERED LIDOCAINE

Buffering lidocaine using sodium bicarbonate raises the pH of the anesthetic solution. In the medical literature, there is evidence that the use of buffered lidocaine results in less pain during the injection.[48,139] In the dental literature, some studies[11,12,34,177,296] found that buffered lidocaine produced less pain on injection and a faster onset of anesthesia. However, other dental studies[274,351,296,155] did not find less pain on injection or a faster onset with buffered lidocaine for IAN block. In symptomatic patients with a diagnosis of pulpal necrosis and associated acute swelling, no significant decrease in pain of infiltrations or significant decrease in the pain of an incision-and-drainage procedure was found when the buffered anesthetic formulation was used.[30,148]

DIPHENHYDRAMINE AS A LOCAL ANESTHETIC AGENT

Diphenhydramine (Benadryl) has been advocated for patients who are allergic to local anesthetics. Multiple studies

have found lower effectiveness as well as more pain and adverse effects with injection of this drug.[355,59] Diphenhydramine should not be used as an alternative to local anesthetics.

Alternative Injection Sites

GOW-GATES AND VAZIRANI-AKINOSI TECHNIQUES

Some clinicians have reported that the Gow-Gates technique[132] has a higher success rate than the conventional IAN block injection,[211,213] but controlled experimental studies have failed to show superiority of the Gow-Gates technique.[8,128,233,329,297] Studies have found no superiority of Gow-Gates for pulpal anesthesia in cases of symptomatic irreversible pulpitis.[4,306,60] A recent study indicated that a combination of Gow-Gates and IAN block injection was significantly superior to the Gow-Gates or IAN blocks alone.[297] The Vazirani-Akinosi technique[9,146,213] also has not been found to be superior to the standard inferior alveolar injection.[146,219,314,329,369] The Vazirani-Akinosi technique is indicated for cases involving a limited mandibular opening to assist with further opening. After using the Vazirani-Akinosi technique, conventional IAN blocks should be given to ensure pulpal anesthesia.

ARTICAINE INFILTRATIONS

Articaine is significantly better than lidocaine for buccal infiltration of the mandibular first molar.[69,171,173,285] However, articaine alone does not predictably provide pulpal anesthesia of the mandibular first molar. There is no difference between 4% articaine with 1:100,000 and 1:200,000 epinephrine for buccal infiltration for pulpal anesthesia.[223]

In anterior teeth, buccal and lingual infiltrations of articaine provide initial pulpal anesthesia, but the anesthesia declines over 60 minutes.[161,256]

Volume and Concentration

Increasing the volume of 2% lidocaine with epinephrine to 3.6 mL (two cartridges) does not increase the incidence of pulpal anesthesia with the IAN block (Fig. 6.6).[105,252,366] Also, no advantage has been seen in using a higher concentration (1:50,000 versus 1:100,000) of epinephrine.[73,341]

Factors in Failure of the Inferior Alveolar Nerve Block

Many theories abound to explain the failure to achieve pulpal anesthesia with an IAN block; these include accessory and cross innervation, injection accuracy, needle beveling, and needle deflection. The mylohyoid nerve is the accessory nerve most often cited as a cause of failure of mandibular anesthesia.[111,366] A controlled clinical trial compared the IAN block alone with a combination of the IAN block and a mylohyoid nerve block using a peripheral nerve stimulator (Fig. 6.7); it was found that the mylohyoid injection did not significantly enhance pulpal anesthesia of the IAN block (Fig. 6.8).[58] Experimentally, cross innervation has been implicated in the failure of anterior pulpal anesthesia. Although this does occur in incisors,[289,368] it plays a very small role in the failure of an IAN block. It has also been theorized that an inaccurate injection contributes to inadequate mandibular anesthesia, but a number of studies have determined that the use of ultrasound, a peripheral nerve stimulator, or radiographs to guide needle placement for IAN blocks did not result in more successful pulpal anesthesia.[26,116,140,309] Needle deflection has also been proposed as a cause of failure with the IAN block.[64,75,156] Several in vitro studies have shown that beveled needles tend to deflect toward the nonbeveled side (i.e., away from the bevel).[10,64,75,156,168,286] In one clinical investigation, however, no significant difference in anesthetic success was seen when the needle bevel was oriented away from the mandibular ramus (so that the needle would deflect toward the mandibular foramen [90% success])

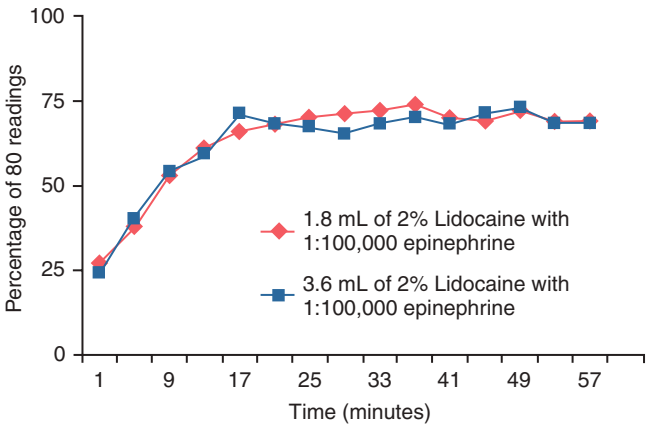


Fig. 6.6 Incidence of first mandibular molar anesthesia: comparison of 1.8 and 3.6 mL of 2% lidocaine with 1:100,000 epinephrine. Results were determined by lack of response to electrical pulp testing at the maximum setting (percentage of 80 readings) across time for 60 minutes. No significant difference between the two volumes was noted.



Fig. 6.7 Injection site for the mylohyoid nerve block.

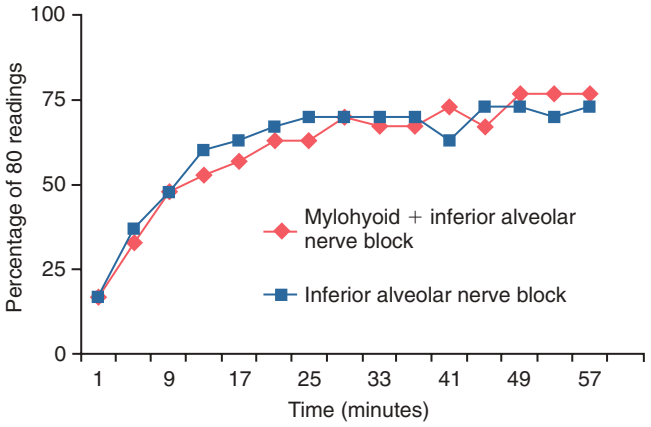


Fig. 6.8 Incidence of first mandibular molar anesthesia: comparison of the combination mylohyoid infiltration plus the inferior alveolar nerve block to the inferior alveolar nerve block alone. Results were determined by lack of response to electrical pulp testing at the maximum setting (percentage of 80 readings) across time for 60 minutes. No significant difference between the two techniques was noted.

compared with the needle bevel oriented toward the mandibular ramus (92% success).[324]

SPEED OF INJECTION AND SUCCESS

A slow IAN block is more successful than a fast injection,[172] but not for patients diagnosed with irreversible pulpitis.[6]

Supplemental Anesthesia for Vital Pulps in the Mandible

INDICATIONS

A supplemental injection is used if the standard injection is not effective. It is useful to repeat an initial injection only if the patient is not exhibiting the classic signs of soft-tissue anesthesia, such as profound lip numbness. Generally, if the classic signs are present, reinjection is not very effective.[175] For example, after the IAN block, the patient develops lip, chin, and tongue numbness and quadrant "deadness" of the teeth. A useful procedure, as previously described, is to pulp test the tooth with a cold refrigerant or an EPT before the cavity preparation is begun.[49,88] If the patient feels pain to cold, a supplemental injection is indicated.

Supplemental injections include the infiltration injection, the IO injection, the intraligament (IL) injection (formerly called the periodontal ligament [PDL] injection), the intraseptal (IS) injection, and the intrapulpal (IP) injection.

INFILTRATIONS

Supplemental Mandibular Buccal, Lingual, or Buccal Plus Lingual Infiltrations of Articaine

Although the infiltration of articaine is effective in restorative dentistry as a supplemental technique (after the IAN block),[138] its use in endodontically involved teeth does not result in predictable pulpal anesthesia.[104,221,257,304,310] A buccal infiltration of lidocaine is also not effective for profound, predictable pulpal anesthesia. An important clinical finding is that an articaine infiltration of the first molar, premolars, and anterior teeth after an IAN block should provide pulpal anesthesia for approximately 1 hour.[138,174,256]

Buccal infiltration alone and buccal plus lingual infiltrations alone (or after an IAN block) do not result in complete pulpal anesthesia.[3,4,7,87,95,273,305] Therefore IO and IL injections are the preferred techniques for supplemental anesthesia, and the IP injection is indicated when the other injections are not successful.

INTRAOSSEOUS ANESTHESIA

The IO injection is a supplemental technique that has been shown to be effective by substantial research and clinical use. It is particularly useful in conjunction with an IAN block when it is likely that supplemental anesthesia will be necessary (e.g., in mandibular second molar teeth).[90,118,135] The IO injection allows placement of a local anesthetic directly into the cancellous bone adjacent to the tooth. When a primary IO injection was compared with an infiltration injection, the IO technique showed a quicker onset and a shorter duration of anesthesia (Fig. 6.9).[254]

The Stabident IO system (Fairfax Dental Inc., Miami, FL) utilizes a slow-speed, handpiece-driven perforator and a solid 27-gauge beveled wire that drills a small hole through the cortical plate (Fig. 6.10). The anesthetic solution is delivered

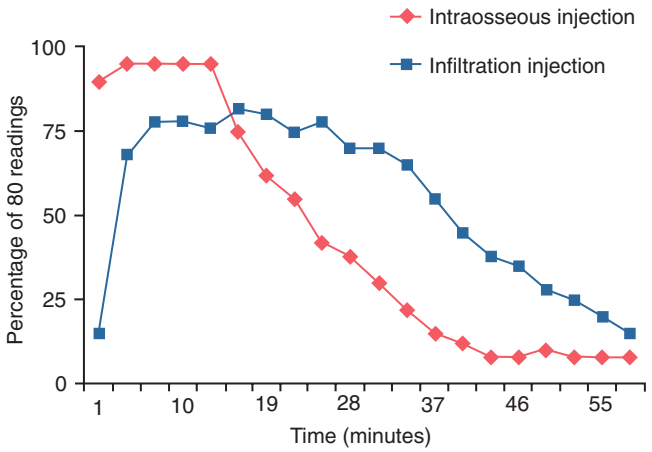


Fig. 6.9 Incidence of anesthesia for intraosseous and infiltration injections. Results were determined by lack of response to electrical pulp testing at the maximum setting (percentage of 80 readings) across time for 60 minutes. The intraosseous injection showed a quicker onset and a shorter duration of anesthesia.

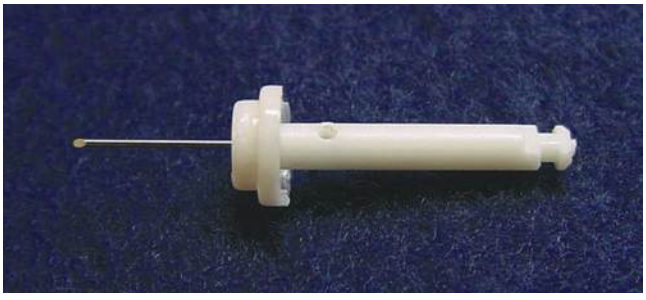


Fig. 6.10 The Stabident perforator, a solid 27-gauge wire with a beveled end that is placed in a slow-speed handpiece.

into cancellous bone through a 27-gauge ultrashort needle placed into the perforation made by the perforator (Fig. 6.11).

The X-Tip IO anesthetic delivery system (Dentsply International, York, PA) consists of an X-tip that separates into two parts, the drill and the guide sleeve (Fig. 6.12). The drill, a special hollow needle, leads the guide sleeve through the cortical plate; it is then separated from the guide sleeve and withdrawn. The remaining guide sleeve is sized to receive a 27-gauge needle for deposition of the anesthetic solution (Fig. 6.13). The guide sleeve is removed after the IO injection is complete.

Another technique available is the TuttleNumbNow (TNN) system. It consists of a Septodont Evolution needle (needle with a secondary scalpel bevel) (Fig. 6.14) and a TNN needle guide that allows the needle to be safely bent by 90 degrees (Fig. 6.15). Instead of a motor-driven perforator, the needle is rotated back and forth 180 degrees as pressure is applied by hand via the anesthetic syringe. The beveled needle is claimed to perforate the cortical bone and allows for delivery of anesthetic into the cancellous bone. It is important to note that this is a system that is intended to deliver anesthetic intraosseously without a separate perforator.

The techniques for IO injection of anesthetic using the Stabident, X-Tip, or TNN system can be reviewed in the systems' instruction manuals or in published papers or bo oks.[61,119,280–282]



Fig. 6.11 The anesthetic solution is delivered to the cancellous bone through a needle placed into the hole made by the perforator.

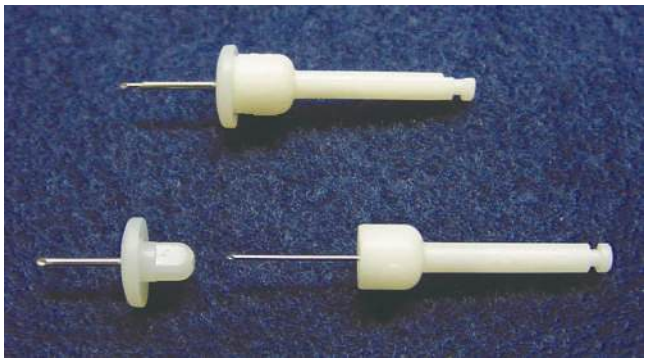


Fig. 6.12 The X-Tip anesthetic delivery system consists of an X-Tip *(top)* that separates into two parts: the drill (a special hollow needle) and the guide sleeve component *(bottom)*.

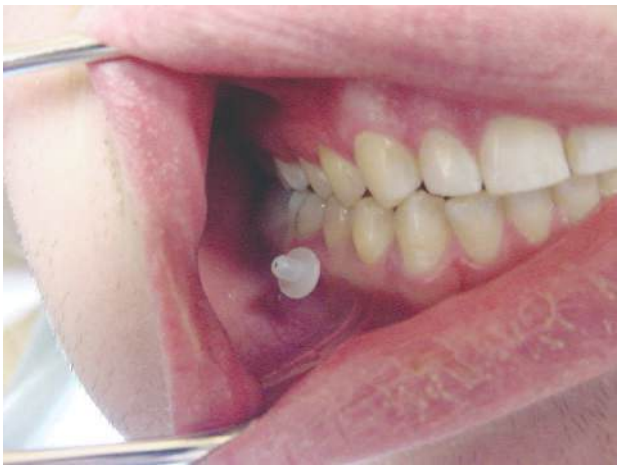


Fig. 6.13 The anesthetic solution is injected through the X-Tip guide sleeve.



Fig. 6.14 Septodont Evolution needle used for the Tuttle Numb Now intraosseous injection technique.



Fig. 6.15 Septodont needle sleeve, which allows safe bending of the needle for the Tuttle Numb Now intraosseous injection technique.

Success and Duration

Supplemental IO injections of lidocaine and mepivacaine with vasoconstrictors allow quick onset and increase the success of the IAN block for approximately 60 minutes (Fig. 6.16).[90,135] The addition of a supplemental IO injection reduced the incidence of slow onset of pulpal anesthesia to zero compared with the IAN block alone (18% incidence).[90] Using 3% mepivacaine plain results in pulpal anesthesia for approximately 30 minutes (Fig. 6.17).[118] Repeating the IO injection—using 1.4 mL of 2% lidocaine with 1:100,000 epinephrine 30 minutes after the initial IO injection—provides an additional 15 to 20 minutes of pulpal anesthesia, similar to the duration of the initial IO injection.[167]

Long-acting anesthetics do not show an extended duration of anesthesia when injected by the IO route.[159,322]

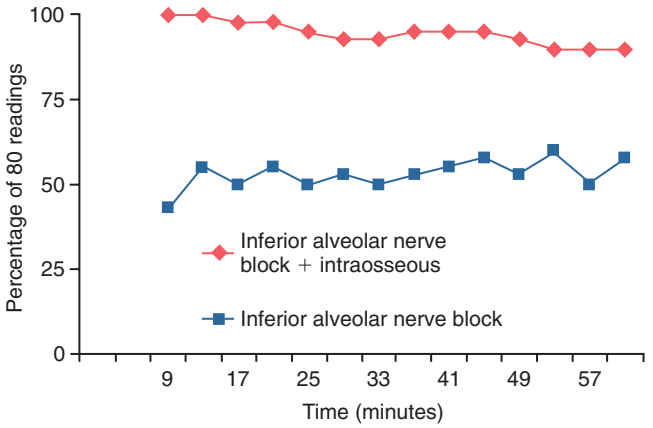


Fig. 6.16 Incidence of first mandibular molar anesthesia: comparison of the combination intraosseous injection of 2% lidocaine with 1:100,000 epinephrine plus the inferior alveolar nerve block to the inferior alveolar nerve block alone. Results were determined by lack of response to electrical pulp testing at the maximum setting (percentage of 80 readings) across time for 60 minutes. The combination technique was significantly better at all postinjection times.

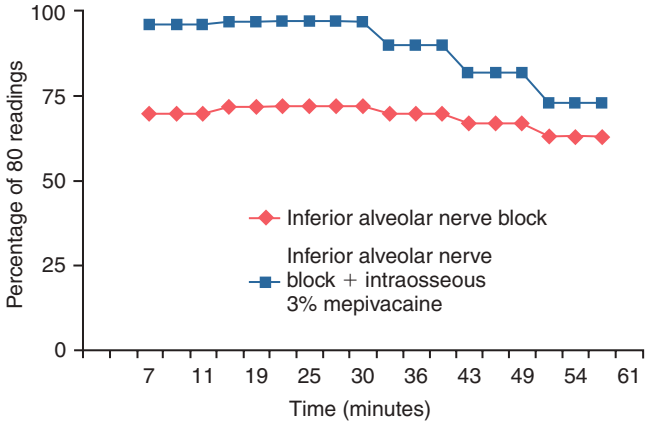


Fig. 6.17 Incidence of first mandibular molar anesthesia: comparison of the combination intraosseous injection with 3% mepivacaine plus the inferior alveolar nerve block to the inferior alveolar nerve block alone. Results were determined by lack of response to electrical pulp testing at the maximum setting (percentage of 80 readings) across time for 60 minutes. The combination technique proved to be significantly better for approximately 30 minutes.

Bupivacaine has cardiotoxic effects[18] and for IO anesthesia offers no advantage, clinically, to 2% lidocaine with epinephrine in terms of efficacy, duration, and heart rate effects. Therefore bupivacaine should not be used for IO anesthesia.

Articaine use in the IO injection has also been found not to provide superior results compared with lidocaine.[27]

Success in Symptomatic Irreversible Pulpitis

High success rates (around 90%) have been reported when the IO injection was used as a supplemental injection with irreversible pulpitis.[27,221,250,253,257] Three percent mepivacaine has an 80% success rate, which increases to 98% with a second IO injection of 3% mepivacaine.[349]

An initial supplemental IO injection of 0.45 to 0.9 mL of 2% lidocaine with 1:100,000 epinephrine was successful in 79% of posterior mandibular teeth.[260] A second IO injection of the remaining cartridge increased the success rate to 91%. Therefore, initially giving a quarter to a half cartridge of 2% lidocaine with 1:100,000 epinephrine was less successful than initially giving a full cartridge.

Although some authors[264,278] have suggested an IO injection alone successfully anesthetizes patients presenting with irreversible pulpitis, it is very doubtful that this would be successful.[27,96,221,250,253,257]

Complications/Contraindications. During an IO injection, if the anesthetic solution flows out of the perforation site (backflow), anesthesia will not occur.[120] Reperforation or use of another perforation site is a practical way to gain access to the cancellous bone in such cases. In fewer than 10% of cases, constricted cancellous spaces may limit the distribution of the anesthetic solution around the apices of the teeth.[61,90,120,135,250,253,279–282,322] In such cases, failure may result even if the anesthetic solution is delivered intraosseously. Reports have shown that approximately 1% of Stabident cases had the metal perforator separate from the plastic shank of the Stabident system during use.[61,90,120,250,280–282] The metal perforator can be easily removed with a hemostat but may require a small gingival flap to access the separated component. Separation usually occurs during a difficult perforation (e.g., dense cortical bone) and may be caused by the wire being heated excessively due to friction, thus causing the plastic hub to melt. No perforator breakage (metal perforator breaking into parts) has been reported in numerous published clinical studies.[61,90,120,250,280–282] However, excessive torqueing of the perforator laterally by an inexperienced operator may result in breakage. When the IO injection is used as a primary injection, pain is experienced about 25% of the time.[61,119,280–282] When the IO injection is used as a supplemental injection, fewer patients experience pain.[90,119,135,280,322] With the Stabident system, fewer than 5% of patients develop swelling or exudate at the site of bone perforation.[61,135,280–282] The X-Tip system may show a higher incidence of postoperative swelling.[119] With both systems, the swelling and/or exudate may persist for several weeks after the injection, but should resolve with time.[63,119,135,280–282] Slow healing of perforation sites may be the result of overheating of the bone caused by pressure during perforation.

With both the Stabident and X-Tip systems, approximately 4% to 15% of patients may report that the tooth "feels high" during chewing.[63,119,135,280–282] This sensation is probably due to an increased awareness to pressure due to soreness in the treated area caused by damage from perforation or inflammation of the bone. The incidence of this effect is lower for the IO injection than that reported for the IL injection (36% to 49%).[72,300] When IO injections are given with the Stabident system, either as a primary or a supplemental technique, most patients report no or only mild pain; approximately 2% to 15% report moderate pain.[63,135,280–282] Less postoperative discomfort is reported for the Stabident IO injection than for IL injection.[300]

One study found that significantly more men experienced postoperative pain with the X-Tip system than with the Stabident system.[119] The authors interpreted these results as being due to the denser, more mineralized bone in the posterior mandible in men and to the fact that the diameter of the X-Tip perforating system is larger than that of the Stabident perforator, meaning the X-Tip system generates more frictional heat during perforation. No clinical research has been published about the TNN system.

Systemic Effects

A transient increase in heart rate has been reported in 46% to 93% of cases involving Stabident and X-Tip IO injection of solutions containing epinephrine or levonordefrin.[61,90,120,135,250,253,280–282,322] Four clinical trials using techniques such as objective electrocardiographic recordings and pulse oximetry have shown that subjects experienced a transient tachycardia (mean increase of 12 to 32 beats/min) after Stabident IO injection of 1.8 mL of 2% lidocaine with 1:100,000 epinephrine, 1.8 mL of 2% mepivacaine with 1:20,000 levonordefrin, or 1.8 mL of 1.5% etidocaine with 1:200,000 epinephrine.[50,135,283,322] Another clinical trial reported transient tachycardia after IO injection, but not with infiltration injection, of 1.8 mL of 2% lidocaine with 1:100,000 epinephrine in the maxillary anterior region.[358] All of these studies showed that the heart rate returned to original baseline readings within 4 minutes in most patients. Therefore IO injection of anesthetic solutions containing vasoconstrictors with either the Stabident or X-Tip system results in a transient tachycardia. No research has been conducted with the TNN system, although similar results would be expected. However, no significant changes in diastolic, systolic, or mean arterial blood pressure has been reported with the IO injection of 2% lidocaine with 1:100,000 epinephrine.[50,283]

Although the patient is likely to notice the transient tachycardia that occurs after Stabident or X-Tip IO injection of 2% lidocaine with 1:100,000 epinephrine, it is generally not clinically significant in healthy patients.[283] The clinical significance, cardiovascular effects, and contraindications to the use of vasoconstrictors in IO injections have been reviewed.[283]

No significant tachycardia occurs when 3% mepivacaine is used for IO anesthesia.[118,283] Clinicians should keep in mind that this anesthetic is an alternative for IO injection in patients whose medical condition or drug therapy suggests caution in the use of solutions containing epinephrine or levonordefrin.[118,283]

Plasma Levels of Lidocaine After Intraosseous Injection. Some authors have warned that administration of a large volume of local anesthetic via an IO injection could lead to overdose reactions.[160] One experimental study using volunteers found that injection of 1.8 mL of 2% lidocaine with 1:100,000 epinephrine produced venous plasma levels of lidocaine that were the same for maxillary anterior IO and infiltration injections (Fig. 6.18).[358] Although systemic concentrations of vasoconstrictors have a short-lived effect on the heart rate, the plasma concentration of lidocaine delivered with IO injection is similar to that delivered via infiltration, so the IO technique should not be considered an intravascular injection with regard to lidocaine. Therefore the precautions for the maximum amount of lidocaine for an infiltration injection would seem to apply to an IO injection.

INTRALIGAMENT ANESTHESIA

The IL injection is another technique that is used if a conventional injection is unsuccessful.[317,344] The technique for the IL injection of anesthesia is reviewed in a number of published papers and textbooks.

Success, Onset, and Duration

For use as a primary injection, IL injections have a reported success rate of about 75% in mandibular posterior teeth, with a duration of pulpal anesthesia of 10 to 15 minutes.[300,352] Success rates are low in anterior teeth.[224,300,352]

For use as a supplemental injection (e.g., when standard techniques have failed to provide adequate anesthesia), good success rates are achieved, but the duration of pulpal anesthesia is approximately only 23 minutes.[54]

The onset of anesthesia is immediate with an IL injection,[72,324,300,317,344,352] which means that no waiting period is required for the anesthesia to take effect. If anesthesia is still not adequate, reinjection may be necessary.

Success in Symptomatic Irreversible Pulpitis

The overall success of supplemental IL injections in achieving pulpal anesthesia for endodontic procedures has been reported to be 50% to 96%.[63,212,317,344] If a first IL injection

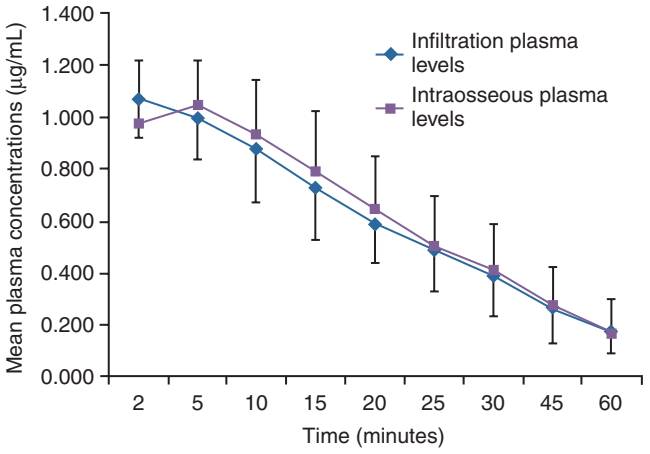


Fig. 6.18 Mean plasma concentrations of lidocaine for the intraosseous and infiltration injection techniques. No statistical differences were seen between the two techniques at any time period.

fails, reinjection results in a success rate of 92%.[344] Similar results have been reported by other investigators.[317] IL injection is not successful in mandibular anterior teeth.[226,352] In addition, supplemental IL injections are not as successful as supplemental IO injections.[175,370]

Mechanism of Action

An IL injection forces anesthetic solutions through the cribriform plate into the cancellous bone around the tooth.[88,113,276,316,346] The primary route of anesthetic is *not* via the PDL, and unlike the IP injection,[329,339] the mechanism of action is not a pressure anesthesia.[92,234] The IL injection should be considered an IO injection.

Back Pressure and Amount of Solution Delivered

Studies have shown that the most important factor for anesthetic success with an IL injection is delivery of anesthetic under *strong back pressure*.[316,344] Pressure is necessary to force the solution into the marrow spaces. Usually about 0.2 mL of solution is deposited with each injection and the use of traditional or pressure syringes. However, the exact amount of anesthetic delivered is not always known because some of the anesthetic solution may escape from the injection site.

Anesthetic Solutions

A vasoconstrictor significantly increases the efficacy of an IL injection.[133,179,190,226,300] Anesthetic solutions with reduced vasoconstrictor concentrations are not very effective with the IL technique.[133,169,179] Articaine is equivalent to lidocaine.[5,25]

Injection Discomfort. When the IL injection is used as a primary injection, needle insertion and injection have the potential to be painful about one third of the time.[234,300,352] The IL injection may be quite painful in maxillary anterior teeth[352] and should not be used in these teeth; infiltration is preferred. The potential for pain with the IL technique is greatly reduced when the injection is used as a supplemental injection after an IAN block.[54]

Selective Anesthesia. Although some authors have reported that the IL injection can be used to differentially diagnose pulpally involved teeth,[209] clinical studies have shown that adjacent teeth can also become anesthetized with the IL injection.[234,300,352] Therefore the IL injection should *not* be used for differential diagnosis.

Postoperative Discomfort

When the IL injection is used as a primary technique, postoperative pain occurs in one third to three quarters of patients, with a duration of 14 hours to 3 days.[72,234,300,317,344,352] The discomfort is related to damage from needle insertion rather than to the pressure of depositing the solution.[72] About one fourth of patients report that their tooth feels "high" in occlusion.[300,352]

Systemic Effects

When a high-pressure syringe was used in dogs, IL injection of solutions containing epinephrine caused cardiovascular responses similar to those seen with intravenous injections.[315] Clinical studies using a high-pressure syringe in human beings found that IL injections of such solutions did not significantly change the heart rate, rhythm, or amplitude or the blood pressure.[46,247] These studies support the conclusion that IL injections do not cause significant changes in heart rate in human beings.

Other Factors

Different needle gauges (25, 27, or 30) are equally effective.[344] Special pressure syringes have been marketed but have not been proven more effective than a standard syringe.[300,344,352]

Safety of the Periodontium and Pulp

Clinical and animal studies have demonstrated the safety of the IL injection technique.[36,112,113,117,208,234,266,300,344,352] Minor transient damage occurs only at the site of needle penetration, but the tissue subsequently undergoes repair. In rare cases, periodontal abscesses and deep pocket formation have occurred after IL injections.[50,352] A small clinical risk of periodontal abscess formation and bone loss exists with this technique, and although these effects are rare, the clinician should be aware of them. Localized areas of root resorption after IL injections have also been reported.[265,284]

IL injections have been shown to be safe in cases of mild to moderate gingival inflammation or incipient periodontitis.[70]

Studies have shown that IL injections have no permanent impact on the pulp.[205,234,267,284,300,352] However, IL injection of a solution containing epinephrine does produce a rapid and prolonged decrease in blood flow to the pulp tissue.[190] Some have suggested that using the IL injection technique during restorative procedures could result in an accumulation of inflammatory mediators in the pulp that could not be effectively removed because of the decreased blood flow.[190] This hypothesis was directly tested, and the IL injection of an anesthetic solution containing a vasoconstrictor in conjunction with a deep cavity preparation did *not* produce a more severe pulpal reaction than in controls (cavity preparation only).[270] It was reported that the depth of the cavity preparation was the biggest factor dictating pulpal responses. IL injections are therefore unlikely to cause pulpal necrosis.

Intraseptal Injection

IS injections are somewhat similar to IL injections in that anesthetic is deposited directly into the interdental septum and the anesthetic gains access to the cancellous bone surrounding the tooth. The injection technique involves placing the needle in the center of the papillary triangle at a 45-degree angle and advancing the needle until it contacts the bony crest.[295,359] The needle may then be pressed into the IS bone and anesthetic delivered with the same back pressure as achieved during an IL injection.

Clinical research has shown the anesthetic success to range from 35% to 90% as a primary injection when used for extractions, restorations, or anesthetic evaluations.[28,31,38] For patients diagnosed with symptomatic irreversible pulpitis, Webster and coworkers[348] reported that

the success of the IS injection as a supplement to the IAN block was only 29%.

Intrapulpal Injection

In about 5% to 10% of mandibular posterior teeth with irreversible pulpitis, supplemental IO or IL injections, even when repeated, do not produce profound pulpal anesthesia and pain is experienced by the patient when the pulp is entered. This is an indication for the IP injection.

The major drawback of the IP injection technique is that the needle and injection are made directly into vital and very responsive pulp tissue; this injection is quite often moderately to severely painful.[253] The endodontic literature offers more efficacious methods of supplemental pulpal anesthesia; therefore the IP injection should be given only after all other supplemental anesthetic techniques have failed. Another disadvantage of the technique is the duration of pulpal anesthesia (15 to 20 minutes). The pulpal tissue must be removed quickly and completely to prevent a recurrence of pain during instrumentation. Also, in order to give the IP injection, the pulp must be exposed to allow direct injection. Anesthetic problems frequently occur before exposure while the clinician is still working in dentin; thus patients may experience pain immediately prior to exposure of the pulp.[221,250,251,253,263,279,310]

The advantage of the IP injection is that it produces profound anesthesia if it is given under back pressure.[29,339] The onset of anesthesia is immediate, and no special syringes or needles are required. The methods for this technique can be found in many endodontic textbooks. As mentioned previously, strong back pressure is the major factor in achieving successful IP anesthesia.[29,339] Simply depositing the anesthetic solution into the chamber will be insufficient because the anesthetic solution will not diffuse throughout the pulp tissue.

Maxillary Anesthesia

Descriptions of conventional techniques for maxillary anesthesia are available for review in numerous articles and textbooks.[165,213] The most commonly used injection for anesthetizing maxillary teeth is infiltration with a cartridge of 2% lidocaine with 1:100,000 epinephrine. Infiltration results in a fairly high incidence of successful pulpal anesthesia (around 87% to 92%).[40,94,134,178,202,220,231,254,261,277,302] However, some patients may not be anesthetized because of individual variations in response to the drug administered, operator differences, and variations of anatomy and tooth position. Pulpal anesthesia usually occurs within 3 to 5 minutes.[40,94,134,178,202,220,231,254,261,277,302] The duration of pulpal anesthesia is a problem with maxillary infiltrations.[40,94,134,178,202,220,236,254,261,277,302] Pulpal anesthesia of the anterior teeth declines after about 30 minutes, with most losing anesthesia by 60 minutes.[40,94,134,178,202,220,236,254,261,277,302] In premolars and first molars, pulpal anesthesia is good until about 40 to 45 minutes and then it starts to decline.[40,94,134,178,202,220,236,254,261,277,302] Additional local anesthetic should be administered depending on the duration of the procedure and the tooth group affected.

Alternative Anesthetic Solutions for Maxillary Infiltrations

PLAIN SOLUTIONS: 3% MEPIVACAINE (CARBOCAINE, POLOCAINE, SCANDONEST) AND 4% PRILOCAINE (CITANEST PLAIN)

Anesthesia duration is shorter with these solutions.[178,220] Therefore these are best used for procedures of short duration (10 to 15 minutes) (Fig. 6.19). They are generally not as safe as anesthetic solutions with vasoconstrictors, especially if *large* volumes are administered. They are rapidly absorbed systemically, thus resulting in excessive plasma concentrations and possible toxic reactions.[213]

4% PRILOCAINE WITH 1:200,000 EPINEPHRINE (CITANEST FORTE), 2% MEPIVACAINE WITH 1:20,000 LEVONORDEFRIN (CARBOCAINE WITH NEO-COBEFRIN), AND 4% ARTICAINE WITH 1:100,000 EPINEPHRINE (SEPTOCAINE, ARTICADENT, ZORCAINE)

These formulations are similar to 2% lidocaine with 1:100,000 epinephrine.[94,178,202]

0.5% BUPIVACAINE WITH EPINEPHRINE (MARCAINE)

Success rates with bupivacaine range from 80% to 95% in the maxillary lateral incisor compared with 50% in the maxillary second premolars.[73,134,187,328] Although bupivacaine can provide long-term anesthesia with the IAN block, it *does not* provide prolonged pulpal anesthesia with maxillary infiltration injection.[74,134,187]

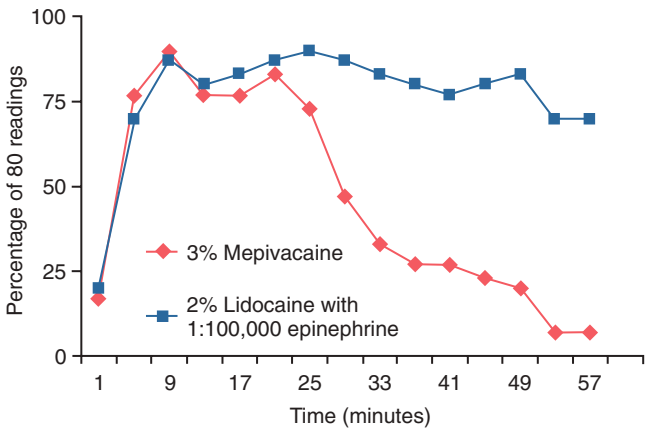


Fig. 6.19 Incidence of first maxillary molar anesthesia: comparison of 3% mepivacaine to 2% lidocaine with 1:100,000 epinephrine. Results were determined by lack of response to electrical pulp testing at the maximum setting (percentage of 80 readings) across time for 60 minutes. The 3% mepivacaine showed a shorter duration of anesthesia than the lidocaine solution.

Extending the Duration of Pulpal Anesthesia for Maxillary Teeth

A two-cartridge volume of 2% lidocaine with epinephrine extends the duration of pulpal anesthesia, but not for 60 minutes.[231] Adding another cartridge of 2% lidocaine with epinephrine at 30 minutes in anterior teeth and 45 minutes in posterior teeth significantly improves the duration of pulpal anesthesia and may be the best way to extend the duration of pulpal anesthesia (Fig. 6.20).[302]

Alternative Maxillary Injection Techniques

For routine restorative and endodontic procedures, posterior superior alveolar (PSA),[209,268] anterior middle superior alveolar (AMSA),[107,108,110,127,203,251] palatal-anterior superior alveolar (P-ASA),[44,109,110,248] infraorbital,[24,176] and second division nerve blocks[39,101] are not advocated. Infiltrations of the area are preferred for pulpal anesthesia.

Kovanaze is an intranasal 3% tetracaine plus 0.05% oxymetazoline formulation designed to provide anesthesia of the maxillary anterior and premolar teeth. Kovanaze is a needle-free anesthetic delivered as an intranasal atomized plume of solution. It was developed as an alternative to needle delivery systems because of the anxiety, fear, and needle sticks associated with traditional systems. Kovanaze has already been studied for its cardiovascular and pharmacokinetic effects[56,152,124] and in restorative procedures in maxillary anterior and posterior teeth.[56,153,152] Studies evaluating pulpal anesthesia are likely to follow to determine its future use in endodontics.

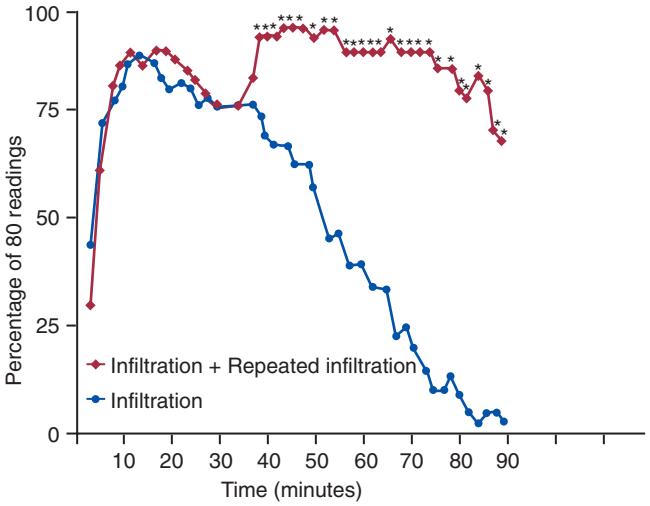


Fig. 6.20 Incidence of maxillary lateral incisor pulpal anesthesia using an initial infiltration and a repeated infiltration 30 minutes later (both infiltrations used 1.8 mL of 2% lidocaine with 1:100,000 epinephrine). Results were determined by lack of response to electrical pulp testing at the maximum setting (percentage of 80 readings). The repeated infiltration injection significantly prolonged the duration of pulpal anesthesia.

Management of Anesthesia in Endodontic Cases

The management of anesthesia in endodontic cases generally includes use of the solutions, injection types, and supplemental anesthesia modalities mentioned throughout this chapter. Anesthesia types and indications will differ depending on pulpal diagnosis. For a thorough review of each clinical scenario with clinical flowcharts, the reader is directed to more in-depth texts on this topic.[277]

Summary and Future Directions for Effective Anesthesia

The emergence of evidence-based dentistry has rapidly drawn interest across all fields of dentistry. Fortunately, the large number of high-quality randomized controlled studies in anesthesia and analgesia provide a wealth of information supporting evidence-based recommendations. The practitioner must focus on high-quality research when clinical decisions are being made.

The prudent practitioner knows that there is no "magic bullet" in pain control. Rather, the clinician must be knowledgeable about the advantages and disadvantages of several anesthetic-vasoconstrictor combinations and the various routes of injection. Clearly, judicious selection of local anesthetics, delivered through multiple sites of injection, is likely to provide predictable anesthesia with minimal exposure to side effects.

ANALGESICS AND THERAPEUTIC RECOMMENDATIONS

Nonnarcotic Analgesics

Management of endodontic pain is multifactorial and directed at reducing the peripheral and central components of hyperalgesia and allodynia through combined endodontic procedures and pharmacotherapy. A major class of drugs for managing endodontic pain comprises the nonnarcotic analgesics, which include both nonsteroidal anti-inflammatory drugs (NSAIDs) and acetaminophen. NSAIDs have been shown to be very effective in managing pain of inflammatory origin and, by virtue of their binding to plasma proteins, actually exhibit increased delivery to inflamed tissue via the extravasation of plasma proteins.[33,82,147] Although these drugs classically are thought to produce analgesia through peripheral mechanisms, the CNS is believed to be an additional site of action.[216,326] NSAIDs inhibit the synthesis of PGs by blocking the enzyme cyclooxygenase (COX), which has two known isoforms, COX-1 and COX-2. Some researchers have proposed that a splice variant of COX-1 (i.e., COX-3) is expressed predominantly in the CNS and is the major site of action of acetaminophen.[52,192,255,301] However, recent studies[151] indicate that the antipyretic and analgesic effects of acetaminophen do not involve inhibition of COX-3; they are more likely exerted through effects of an active metabolite on CNS

cannabinoid receptors,[14] and this metabolite appears to act by blocking a calcium channel ($Ca_V3.2$).[188]

Numerous NSAIDs are available for the management of pain and inflammation (Table 6.4). Ibuprofen is generally considered the prototype of contemporary NSAIDs and has a well-documented efficacy and safety profile.[80] New formulations of ibuprofen include ibuprofen sodium dihydrate, which is an effective analgesic and has a faster onset of action than ibuprofen acid. Other NSAIDs may offer certain advantages over ibuprofen.[327] For example, etodolac (i.e., Lodine) causes less gastrointestinal (GI) irritation,[16] and ketoprofen (i.e., Orudis) has been shown in some studies to be somewhat more analgesic than ibuprofen.[67] An intranasal formulation of ketorolac tromethamine (Sprix, Regency Therapeutics, Shirley, NY) is now available and provides significant pain relief within 30 minutes of administration in patients with endodontic pain.[337] Recent studies suggest that in addition to inhibiting COX, ketorolac and diclofenac inhibit peripheral *N*-methyl-D-aspartate (NMDA) receptors, which may contribute to their analgesic effects.[47,85] The advantages of NSAIDs include their well-established analgesic efficacy for inflammatory pain. Many of the NSAIDs listed in Table 6.4 have been shown to be more effective than traditional acetaminophen and opioid combinations such as acetaminophen with codeine.[66,80,334]

Table 6.4 Summary of Selected Nonnarcotic Analgesics

Analgesic	Trade Name	Dose Range (mg)	Daily Dose (mg)
Acetaminophen	Tylenol and others	325–650	4000
Aspirin	Many	325–1000	4000
Diclofenac potassium	Cataflam	50–100	150–200
Diflunisal	Dolobid	250–1000	1500
Etodolac	Lodine	200–400	1200
Fenoprofen	Nalfon	200	1200
Flurbiprofen	Ansaid	50–100	200–300
Ibuprofen acid	Motrin et al.	200–800	2400 (Rx)
Ibuprofen sodium dihydrate	Fast-acting Advil	256–512	1536
Ketoprofen	Orudis	25–75	300 (Rx)
Ketorolac*	Toradol Sprix	30–60 (oral) 31.5	60 126 mg
Naproxen	Naprosyn	250–500	1500
Naproxen Na	Anaprox and others	220–550	1650 (Rx)

*A new package insert for ketorolac tablets includes the instructions that the drug should be used only as a transition from injectable ketorolac and for no longer than 5 days. The package insert for intranasal ketorolac states that for patients 65 years of age or older, renally impaired patients, and patients weighing less than 50 kg, the dose should be limited to 15.75 mg and the maximum daily dose should be limited to 63 mg.

Note: Cyclooxygenase-2 (COX-2) inhibitors are not included (see text).

Rx, Prescription strength

Modified from Cooper SA: Treating acute dental pain, *Postgrad Dent* 2:7, 1995.

A 2002 paper represents the first systematic review comparing all endodontic pain studies evaluating oral NSAIDs.[158] The authors concluded that NSAIDs combined with other drugs (e.g., flurbiprofen with tramadol[86]) or pretreatment and posttreatment application of NSAIDs provides effective pain control. Table 6.5 lists the results of a large ongoing systematic review of the relative efficacy of analgesics in inflammatory pain conditions. Importantly, the data are generated based on postoperative patients having moderate to severe pain, and the number needed to treat (NNT) is based on the relative superiority of the analgesic over placebo for producing a 50% relief in pain. These data, therefore, constitute important, clinically relevant information for clinicians who want to compare the relative efficacy of posttreatment analgesics. Of course other issues, such as the potential adverse effects of drugs and the patient's medical history, must be considered when a treatment plan for postendodontic pain is developed.

The introduction of selective inhibitors of COX-2 offered the potential for both analgesic and anti-inflammatory benefits and reduced GI irritation.[81,189] However, most selective COX-2 inhibitors have been withdrawn from the market due to prothrombic adverse effects, and the only coxib still available, celecoxib, has not received FDA approval for the treatment of acute inflammatory pain. Concern has been raised that the COX-2 inhibitors may also cause at least some GI irritation in patients with preexisting GI disease.[342]

Another major concern centers on the recognized prothrombic adverse effects of the COX-2 inhibitors. Two meta-analyses have examined the cardiovascular (CV) safety of traditional NSAIDs and COX-2 inhibitors. Kearney et al.[182] conducted a meta-analysis of 138 randomized trials, and McGettigan and Henry[224] conducted a meta-analysis of 23 controlled observational studies.[182,224] Kearney et al.[182] estimated a relative risk for CV events associated with COX-2 to be 1.42 (95% CI, 1.64 to 2.91). Naproxen was found to have no significant adverse effects on the CV system in either meta-analysis. Diclofenac (Voltaren) is a relatively COX-2–selective drug and seems to have a degree of COX-2 selectivity similar to that of celecoxib. Diclofenac was associated with increased CV events. In the randomized trial analysis, there was an increase in CV risk with high-dose ibuprofen. Based on the available data, the FDA has requested that manufacturers of all prescription products containing nonselective NSAIDs revise their product labeling to include (1) a boxed warning regarding the potential serious adverse CV events and the serious, potentially life-threatening GI adverse events associated with the use of this class of drugs; (2) a contraindication to use in patients who have recently undergone coronary artery bypass surgery; and (3) a medication guide for patients regarding the potential for CV and GI adverse events associated with the use of this class of drugs. The available data do not suggest an increased risk of serious CV events for the short-term, low-dose use of NSAIDs available over the counter, but the FDA has requested changes to the label to better inform consumers about the safe use of these products. Given this situation and reasonable alternative NSAIDs, we recommend against the use of COX-2 inhibitors for the treatment of patients with routine endodontic pain.

Table 6.5 Oxford League Table of Analgesic Efficacy*

Analgesic†	Number of Patients in Comparison	Percentage With at Least 50% Pain Relief	Number Needed to Treat	Lower Confidence Interval	Higher Confidence Interval
Ibuprofen (800 mg)	76	100	1.6	1.3	2.2
Ketorolac (60 mg) (IM)	116	56	1.8	1.5	2.3
Diclofenac (100 mg)	548	69	1.8	1.6	2.1
Oxycodone IR (5 mg) + acetaminophen (500 mg)	150	60	2.2	1.7	3.2
Diclofenac (50 mg)	738	63	2.3	2	2.7
Naproxen (440 mg)	257	50	2.3	2	2.9
Oxycodone IR (15 mg)	60	73	2.3	1.5	4.9
Ibuprofen (600 mg)	203	79	2.4	2	4.2
Ibuprofen (400 mg)	5456	55	2.5	2.4	2.7
Aspirin (1200 mg)	279	61	2.4	1.9	3.2
Oxycodone IR (10 mg)	315	66	2.6	2	3.5
Ketorolac (10 mg) + acetaminophen (650 mg)	790	50	2.6	2.3	3.1
Ibuprofen (200 mg)	3248	48	2.7	2.5	2.9
Naproxen (500/550)	784	52	2.7	2.3	3.3
Diclofenac (50 mg)	1296	57	2.7	2.4	3.1
Diclofenac (25 mg)	204	54	2.8	2.1	4.3
Demerol (100 mg) (IM)	364	54	2.9	2.3	3.9
Tramadol (150 mg)	561	48	2.9	2.4	3.6
Morphine (10 mg) (IM)	946	50	2.9	2.6	3.6
Naproxen (500/550 mg)	784	52	2.7	2.3	3.3
Naproxen (220/250 mg)	202	45	3.4	2.4	5.8
Ketorolac (30 mg) (IM)	359	53	3.4	2.5	4.9
Acetaminophen (500 mg)	561	61	3.5	2.2	13.3
Acetaminophen (600/650 mg) + codeine (60 mg)	1123	42	4.2	3.4	5.3
Acetaminophen (650 mg) + dextropropoxyphene (65 mg hydrochloride or 100 mg napsylate)	963	38	4.4	3.5	5.6
Aspirin (600/650 mg)	5061	38	4.4	4	4.9
Acetaminophen (600/650 mg)	1886	38	4.6	3.9	5.5
Tramadol (100 mg)	882	30	4.8	3.8	6.1
Tramadol (75 mg)	563	32	5.3	3.9	8.2
Aspirin (650 mg) + codeine (60 mg)	598	25	5.3	4.1	7.4
Oxycodone IR (5 mg) + acetaminophen (325 mg)	149	24	5.5	3.4	
Ketorolac (10 mg) (IM)	142	48	5.7	3	53
Acetaminophen (300 mg) + codeine (30 mg)	379	26	5.7	4	9.8
Tramadol (50 mg)	770	19	8.3	6	13
Codeine (60 mg)	1305	15	16.7	11	48
Placebo	>10,000	18	N/A	N/A	N/A

*Analgesics are listed in descending order from most to least effective based on the NNT. The NNT reflects the superiority of an analgesic over a placebo treatment; therefore an analgesic with a lower NNT has greater efficacy than an analgesic with a higher NNT. The NNT is calculated for the proportion of patients with at least 50% pain relief over 4 to 6 hours compared with placebo in randomized, double-blind, single-dose studies in patients with moderate to severe pain. The 95% confidence interval contains the upper and lower estimates of the NNT with a 95% chance of accuracy.

†Drugs were given orally except where noted.

IM, Intramuscular; *IR,* immediate release; *NA*, not applicable

Modified from http://www.bandolier.org.uk/booth/painpag/Acutrev/Analgesics/lftab.html.

LIMITATIONS AND DRUG INTERACTIONS

Clinicians should educate themselves not only on the efficacy of the nonnarcotic analgesics but also on their limitations and interactions with other drugs.[45] For example, NSAIDs exhibit an analgesic ceiling that limits the maximal level of analgesia; they also induce side effects, including those affecting the GI system (3% to 11% incidence) and the CNS (1% to 9% incidence of dizziness and headache). NSAIDs are contraindicated in patients with ulcers and aspirin hypersensitivity.[199,357] They are also associated with severe GI complications, and the risk of adverse effects increases with an increasing lifetime-accumulated dose of these drugs and the concurrent intake of aspirin, steroids, or coumadin.[15,83,199,357] A strategy for avoiding GI bleeding associated with NSAIDs is to use a proton pump inhibitor (PPI). A combination of a PPI (esomeprazole magnesium) with naproxen (Vimovo, Patheon/Astrozenica, Wilmington, DE) is now available. Although this is a cost-effective approach, it does not protect the lower GI tract.[321] Another approach is to use a histamine receptor H_2 antagonist in combination with an NSAID. The combination of famotidine (an H_2-receptor antagonist) and ibuprofen (Duexis, Horizon Pharma, Northbrook, IL) was recently approved by the FDA. Also, as mentioned earlier, there is an increased CV risk with some NSAIDs, including ibuprofen.

The NSAIDs have been reported to interact with a number of other drugs (Table 6.6). Acetaminophen and opioid combination drugs are alternatives for patients unable to take NSAIDs.[65] Further information is available on the pharmacology and adverse effects of this important class of drugs.[45,63,83,115,364] Other resources are also available for the evaluation of drug interactions, including internet drug search engines such as *rxlist.com*, *Epocrates.com*, and *Endodontics.UTHSCSA.edu*.

Table 6.6 Summary of Selected Drug Interactions With Nonsteroidal Anti-inflammatory Drugs

Drug	Possible Effect
Angiotensin-converting enzyme (ACE) inhibitors	Reduced antihypertensive effectiveness of captopril (and especially indomethacin)
Anticoagulants	Increase in prothrombin time or bleeding with anticoagulants (e.g., coumarins)
Beta blockers	Reduced antihypertensive effects (e.g., propranolol, atenolol, pindolol)
Cyclosporine	Increased risk of nephrotoxicity
Digoxin	Increase in serum digoxin levels (especially ibuprofen, indomethacin)
Dipyridamole	Increased water retention (especially indomethacin)
Hydantoins	Increased serum levels of phenytoin
Lithium	Increased serum levels of lithium
Loop diuretics	Reduced effectiveness of loop diuretics (e.g., furosemide, bumetanide)
Methotrexate	Increased risk of toxicity (e.g., stomatitis, bone marrow suppression)
Penicillamine	Increased bioavailability (especially indomethacin)
Sympathomimetics	Increased blood pressure (especially indomethacin with phenylpropanolamine)
Thiazide diuretics	Reduced antihypertensive effectiveness

Data from Facts and Comparisons: *Drug facts and comparisons,* ed 54, St Louis, 2000, Facts and Comparisons; Gage T, Pickett F: *Mosby's dental drug reference,* ed 5, St Louis, 2000, Mosby; and Wynn R, Meiller T, Crossley H: *Drug information handbook for dentistry,* Hudson, OH, 2000, Lexi-Comp.

ACETAMINOPHEN

Acetaminophen (*N*-acetyl *p*-aminophenol) is one of the most commonly used analgesic and antipyretic drugs. Acetaminophen alone is not comparable with ibuprofen in relieving moderate to severe pain; however, a combination of ibuprofen and acetaminophen provides greater pain relief than either drug alone, as evidenced by large systematic reviews.[238] In a randomized double-blind placebo-controlled study, patients who had undergone pulpectomy were administered a single dose of either a combination of acetaminophen (1000 mg) and ibuprofen (600 mg) or ibuprofen (600 mg) alone. The combination of the two analgesics provided greater pain relief in the immediate postoperative period (8 hours) than ibuprofen (600 mg) alone.[230]

Acetaminophen is one of the most common drugs found in combination products for the relief of pain and symptoms of cold or flu. It is considered safe when taken at normal doses; in higher doses, however, acetaminophen causes liver toxicity and is associated with nearly half of the cases of acute liver failure in the United States.[201] Most acetaminophen is conjugated in the liver to form inactive metabolites. A small portion is metabolized by the cytochrome P450 system to form *N*-acetyl-*p*-benzoquinone imine (NAPQI), which is very toxic but is generally detoxified by glutathione and converted into nontoxic compounds. Large doses of acetaminophen saturate the main route of metabolism, causing more acetaminophen to be converted to NAPQI. Liver injury occurs once glutathione becomes depleted and NAPQI is allowed to accumulate. To minimize this risk, it has been recommended that healthy adults not take more than 3 g (3000 mg) of acetaminophen in a 24-hour period (*www.tylenolprofessional.com/extra-strength-tylenol-dosage-faq.html*). The FDA requires a boxed warning to be added to the label of all oral prescription drug products that contain acetaminophen. (A boxed warning, the strongest warning that the FDA requires, indicates that the drug carries a significant risk of serious adverse effects.)

Opioid Analgesics

Opioids are potent analgesics and are often used in dentistry in combination with acetaminophen, aspirin, or ibuprofen. Most clinically available opioids activate mu opioid receptors at several important sites in the brain and on afferent neurons. Studies indicate that they activate peripheral opioid receptors in dental pulp and that IL injection of morphine significantly reduces pain in endodontic patients and in other inflammatory pain states.[84,97,143] Gender-dependent differences appear to exist in response to at least

the kappa opioid agonists.[122] In a randomized controlled clinical trial, women taking a combination of pentazocine and naloxone had significantly less postoperative endodontic pain than men taking the same medications.[294]

Although opioids are effective as analgesics for moderate to severe pain, it is important to remember that they have no anti-inflammatory effects and as such will have no effect on the inflamed periapical tissues. The current opioid epidemic in the United States highlights the risks associated with even a single prescription of opioids—that is, opioid diversion, their misuse, and the development of opioid use disorders. Adverse effects associated with these drugs include nausea, emesis, dizziness, drowsiness, and the potential for respiratory depression and constipation. Because the dosage is limited by side effects, these medications are almost always used in combination drugs to manage dental pain. A combination formulation is preferred because it permits a lower dose of the opioid, thereby reducing side effects (Table 6.7).

Codeine is often considered the prototype opioid for orally available combination drugs. Most studies have found that the 60-mg dose of codeine (the amount in two tablets of Tylenol with codeine no. 3) produces significantly more analgesia than placebo, although it often produces less analgesia than either aspirin 650 mg or acetaminophen 600 mg.[65,66] In general, patients taking only 30 mg of codeine report about as much analgesia as those taking a placebo.[22,334] Table 6.8 provides comparable doses of other opioids equivalent to 60 mg of codeine.

Corticosteroids

Posttreatment pain or a flare-up after endodontic treatment can be attributed to inflammation, infection, or both in the periradicular tissues. Establishing patency and subsequently debriding and shaping the root canal system may irritate the periradicular tissues and inadvertently introduce bacteria, bacterial products, necrotic pulp tissue, or caustic irrigating solution through apical foramina.

In response to this irritation, inflammatory mediators (e.g., PGs, leukotrienes, bradykinin, platelet-activating factor, substance P) are released into the tissues surrounding the apical area of the tooth. As a result, pain fibers are directly stimulated or sensitized, and an increase in vascular dilation and permeability results in edema and increased interstitial tissue pressure.

Glucocorticosteroids are known to reduce the acute inflammatory response by suppressing vasodilation, the migration of polymorphonuclear (PMN) leukocytes, and phagocytosis and by inhibiting the formation of arachidonic acid from neutrophil and macrophage cell membrane phospholipids, thus blocking the COX and lipoxygenase pathways and respective synthesis of PGs and leukotrienes. It is not surprising that a number of investigations have evaluated the efficacy of corticosteroids (administered via either intracanal or systemic routes) in the prevention or control of postoperative endodontic pain or flare-ups.[217]

Table 6.7 Selected Opioid Combination Analgesic Drugs

Formulation	Trade Name*	Possible Rx
APAP 300 mg and codeine 30 mg	Tylenol with codeine no. 3	2 tabs q4h
APAP 500 mg and hydrocodone 5 mg	Vicodin, Lortab 5/500	1–2 tabs q6h
APAP 325 mg and oxycodone 5 mg	Percocet	1 tab q6h
APAP 500 mg and oxycodone 5 mg	Tylox	1 tab q6h
ASA 325 mg and codeine 30 mg	Empirin with codeine no. 3	2 tabs q4h
ASA 325 mg and oxycodone 5 mg	Percodan	1 tab q6h

*Several generics are available for most formulations.
APAP, Acetaminophen; *ASA*, acetylsalicylic acid (aspirin); *Rx*, prescription.

Table 6.8 Analgesic Doses of Representative Opioids

Opioid	Dose Equivalent to Codeine 60 mg
Codeine	60 mg
Oxycodone	5–6 mg
Hydrocodone	10 mg
Dihydrocodeine	60 mg
Propoxyphene HCl	102 mg
Propoxyphene-N	146 mg
Meperidine	90 mg
Tramadol	50 mg

HCl, hydrochloride; *N*, napsylate
Modified from Troullos E, Freeman R, Dionne RA: The scientific basis for analgesic use in dentistry, *Anesth Prog* 33:123, 1986.

INTRACANAL ADMINISTRATION

Several studies have evaluated the intracanal administration of steroids.[51,243,333] A large-scale clinical trial of 223 patients reported significantly less posttreatment pain in patients after intracanal administration of Ledermix (triamcinolone acetonide and demeclocycline HCl) compared with either calcium hydroxide or no intracanal dressing.[93] Intracanal steroids appear to have significant effects for reducing postoperative pain.[287]

SYSTEMIC ADMINISTRATION

Some studies have evaluated the systemic route of administration of corticosteroids on posttreatment pain or flare-ups.[181,204,218]

In a double-blind placebo-controlled study, patients with irreversible pulpitis were given 4 mg of dexamethasone or placebo by means of a supraperiosteal injection at the apex of the treated tooth after pulpectomy.[229] This is an injection technique that most clinicians would be familiar with (as opposed to intramuscular injection). Posttreatment pain was significantly reduced in the steroid group during the first 24 hours. There was no difference at 48 hours.

Yet another study evaluated the effect of IO injection of methylprednisolone or placebo in patients with irreversible pulpitis. Highly significant pain reduction in the steroid group was maintained for 7 days after a single injection.[123]

Animal studies have histologically evaluated the anti-inflammatory effects of corticosteroids on inflamed periradicular tissues. In one study, after an acute inflammatory reaction was induced in the molar teeth of rats by overextending endodontic instruments, sterile saline or dexamethasone was infiltrated supraperiosteally into the buccal vestibule adjacent to the treated teeth. Dexamethasone significantly reduced the number of neutrophils present and thus had an anti-inflammatory effect on the periradicular tissues of the treated teeth.[246]

Other studies of systemic administration have evaluated the effect of oral administration of corticosteroids on the incidence and severity of posttreatment endodontic pain. In a double-blind controlled clinical trial, 50 patients randomly received either 0.75 mg of dexamethasone or a placebo tablet orally after initial endodontic treatment.[196] Oral dexamethasone significantly reduced posttreatment pain after 8 and 24 hours compared with placebo. A follow-up study evaluated the effect a larger oral dose of dexamethasone (i.e., 12 mg given every 4 hours) on the severity of posttreatment endodontic pain.[125] Results showed that dexamethasone was effective in reducing posttreatment endodontic pain up to 8 hours after the treatment was completed; no effect on the severity of pain was seen at 24 and 48 hours after treatment. In a double-blind study comparing etodolac and dexamethasone, the perioperative administration of oral dexamethasone was similar to that of etodolac in reducing pain after endodontic surgery. In a randomized clinical trial,[206] 40 patients received a single dose of prednisolone (30 mg) or placebo 30 minutes before initiation of endodontic treatment. Prednisolone significantly reduced posttreatment pain at 6, 12, and 24 hours compared to placebo.[164]

Collectively, these studies on systemic steroid administration indicate that corticosteroids reduce the severity of posttreatment endodontic pain compared with placebo treatment. However, given the relative safety/efficacy relationship between steroids and NSAIDs, most investigators choose an NSAID as the drug of first choice for postoperative pain control.

Of note is an interesting study reporting that 17.7% of patients on steroid therapy developed dentin hypersensitivity–like pain.[308] A positive correlation was noted between the steroid dose and pain scores. Discontinuation of the steroids resolved the pain in all the study subjects.

Antibiotics

Because bacteria are involved in endodontic cases with apical periodontitis, the incidence of a posttreatment infection or flare-up is a concern to clinicians providing endodontic treatment. Prescribing an antibiotic to prevent such an occurrence might make sense, but the use of antibiotics is controversial for several reasons.[102] First, the overprescribing of antibiotics, especially when these drugs are not indicated, has led to increased bacterial resistance and patient sensitization. Second, antibiotics have been mistakenly prescribed for patients with severe pain who have a vital tooth (i.e., when bacteria are unlikely to be a causative factor in periradicular pain).[367] Third, even when bacteria are likely to be present, data from controlled clinical trials provide little or no support for the hypothesis that antibiotics reduce pain.[183]

A series of clinical studies evaluated the efficacy of prophylactically administered systemic antibiotics for preventing posttreatment endodontic flare-ups. Working on the premise that the incidence of infectious flare-ups after endodontic treatment is 15%, Morse et al.[242] randomly prescribed a prophylactic dose of either penicillin or erythromycin to patients after endodontic treatment of teeth with a diagnosis of necrotic pulp and chronic periradicular periodontitis (no placebo was used). The results showed that the overall incidence of flare-ups was 2.2%, with no difference between penicillin and erythromycin. Similar results were obtained in a study in which dental students (instead of private practitioners) provided the endodontic treatment.[1] Outcomes found a 2.6% incidence of flare-ups, with no statistically significant differences between penicillin and erythromycin. However, neither the original nor the follow-up study was a randomized placebo-controlled clinical trial. This observation appears to be highly significant for clinical recommendations because in general, randomized controlled studies fail to detect any analgesic benefit from antibiotics, whereas open-label or historical control studies often report profound effects.[102]

To determine whether the timing of administration of an antibiotic altered the occurrence of flare-ups and pain unassociated with flare-ups, researchers analyzed components of two separate prospective studies of patients undergoing endodontic treatment for teeth with necrotic pulps and chronic periradicular periodontitis. In the first study, prophylactic penicillin was provided; in the second study, patients were instructed to take penicillin (or erythromycin if they were allergic to penicillin) at the first sign of swelling.[241,242] The authors concluded that prophylactic administration of antibiotics is preferable to having the patient take antibiotics at the first sign of an infection.

In a multicenter two-part clinical study, 588 consecutive patients received one of nine medications or placebos and were monitored for 72 hours after treatment.[331] The results showed that ibuprofen, ketoprofen, erythromycin, penicillin, and penicillin plus methylprednisolone significantly reduced the severity of pain within the first 48 hours after treatment. The second part of the study then evaluated the incidence of posttreatment pain after obturation of the same teeth in the first phase of the study.[331] Although only 411 of the original 588 patients participated in this phase, they were randomly given the same medications or placebo after completion of the obturation appointment. The incidence of posttreatment pain was lower after obturation (5.83%) than after cleaning and shaping (21.76%) of the root canal system, and no significant difference was found in the effectiveness of the various medications and placebo in controlling posttreatment pain after obturation.

Walton and Chiappinelli[345] were concerned that previous studies were uncontrolled, retrospective, or carried out on different patient groups at different times and with different treatment modalities. They conducted a randomized prospective double-blind clinical trial to test the hypothesis that an antibiotic (e.g., penicillin) can prevent a posttreatment endodontic flare-up.[345] Eighty patients with teeth that had necrotic pulp and chronic periradicular periodontitis were randomly divided into three groups. In the first two

groups, either penicillin or a placebo was administered 1 hour before and 6 hours after the individual appointments on a double-blind basis. Upon completion of the individual appointments—which included debridement, shaping, and possibly obturation of the root canal system—the patients completed questionnaires at 4, 8, 12, 24, and 48 hours. No significant difference was found among the three groups in the incidence of flare-ups, pain, or swelling. The authors concluded that the use of prophylactic penicillin offers no benefit for postoperative pain or flare-up, and prophylactic administration of penicillin should not be used routinely in patients undergoing endodontic treatment of necrotic teeth and chronic periradicular periodontitis.

In another randomized prospective placebo-controlled clinical study, a group of investigators examined whether supplemental penicillin reduced symptoms or shortened the course of recovery of emergency patients diagnosed with necrotic pulp and acute apical abscess.[103] Patients were randomly given penicillin, a placebo, or no medication. Using a visual analog scale, the subjects then evaluated their postoperative pain and swelling for up to 72 hours. No significant difference was found among the three groups. Recovery occurred as a result of endodontic treatment alone.

It is well recognized that antibiotics may be indicated for the management of infections of endodontic origin. However, a review of the available literature indicates that prophylactic use is contraindicated in immunocompetent patients with no systemic signs of infection and with swelling localized to the vestibule. Controlled clinical studies indicate that antibiotics offer little or no benefit for pain reduction under these circumstances, but they may be indicated for immunocompromised patients and for patients with systemic signs and symptoms of an infection or an infection that has spread into the fascial spaces of the head and neck.

Pain Management Strategies

When pain in an individual patient is being managed, the skilled clinician must customize the treatment plan, balancing the general principles of endodontics, mechanisms of hyperalgesia, and pain management strategies with the particular factors of the individual patient (e.g., medical history, concurrent medications).[147,184,185,291,313]

Effective management of endodontic pain starts with the "three Ds": *diagnosis*, *definitive* dental treatment, and *drugs* (Box 6.1). Comprehensive reviews on *diagnosis* and *definitive* dental treatment (e.g., incision and drainage, pulpectomy) are provided elsewhere in this text. As described earlier in this chapter, the management of endodontic pain should focus on the removal of peripheral mechanisms of hyperalgesia and allodynia (see Box 6.1). This generally requires treatment that removes and reduces causative factors (e.g., bacterial and immunologic factors). For example, both pulpotomy and pulpectomy have been associated with substantial reduction in patient reports of pain compared to pretreatment pain levels.[86,150,222,263] However, pharmacotherapy is often required to reduce continued nociceptor input (e.g., NSAIDs, local anesthetics) and suppress central hyperalgesia (e.g., NSAIDs, opioids).

Box 6.1 Considerations for Effective "Three D" Pain Control

1. Diagnosis
2. Definitive dental treatment
3. Drugs
   - Pretreatment with nonsteroidal anti-inflammatory drugs or acetaminophen when appropriate.
   - Use of long-acting local anesthetics when indicated.
   - Use of a flexible prescription plan.
   - Prescription "by the clock" rather than as needed.

PRETREATMENT

Treatment with an NSAID before a procedure has been shown to have a significant benefit in many studies[80,162] but not all.[245] The rationale for pretreatment is to block the development of hyperalgesia by reducing the input from peripheral nociceptors. For patients who cannot take NSAIDs, pretreatment with acetaminophen has also been shown to reduce postoperative pain.[236] Patients can be pretreated 30 minutes before the procedure with either an NSAID (e.g., ibuprofen 400 mg or flurbiprofen 100 mg) or acetaminophen 1000 mg.[86,162,236]

LONG-ACTING LOCAL ANESTHETICS

A second pharmacologic approach for pain management is the use of long-acting local anesthetics, such as bupivacaine and ropivacaine. Clinical trials indicate that long-acting local anesthetics not only provide anesthesia during the procedure but also significantly delay the onset of posttreatment pain compared with local anesthetics that contain lidocaine.[80,130,131,166] Administration of long-acting local anesthetics by block injection has been shown to reduce posttreatment pain for 2 to 7 days after the oral procedure[130,131,166] because an afferent barrage of nociceptors can induce central hyperalgesia.[360–362] The analgesic benefit of long-acting local anesthetics is more prominent with block injections than infiltration injections, but the clinician should also be aware of adverse effects attributed to these agents.[18,235]

FLEXIBLE PLAN

A third pharmacologic approach is the use of a flexible plan for prescribing analgesics (Fig. 6.21).[13,66,142,144,147,184,186,334] A flexible prescription plan serves to minimize both postoperative pain and side effects. Given these goals, the clinician's strategy is twofold: (1) to achieve a maximally effective dose of the nonnarcotic analgesic (either an NSAID or acetaminophen for patients who cannot take NSAIDs) and (2) in the rare cases in which the patient still has moderate to severe pain, to consider adding drugs that increase the NSAID's analgesia. Because of its predictive value, the presence of preoperative pain or mechanical allodynia may be an indicator that such NSAID combinations should be considered.

Most studies report that combining an NSAID with acetaminophen 1000 mg alone (i.e., no opioid) produces nearly twice the analgesic response as just the NSAID.[37,68,230,350]

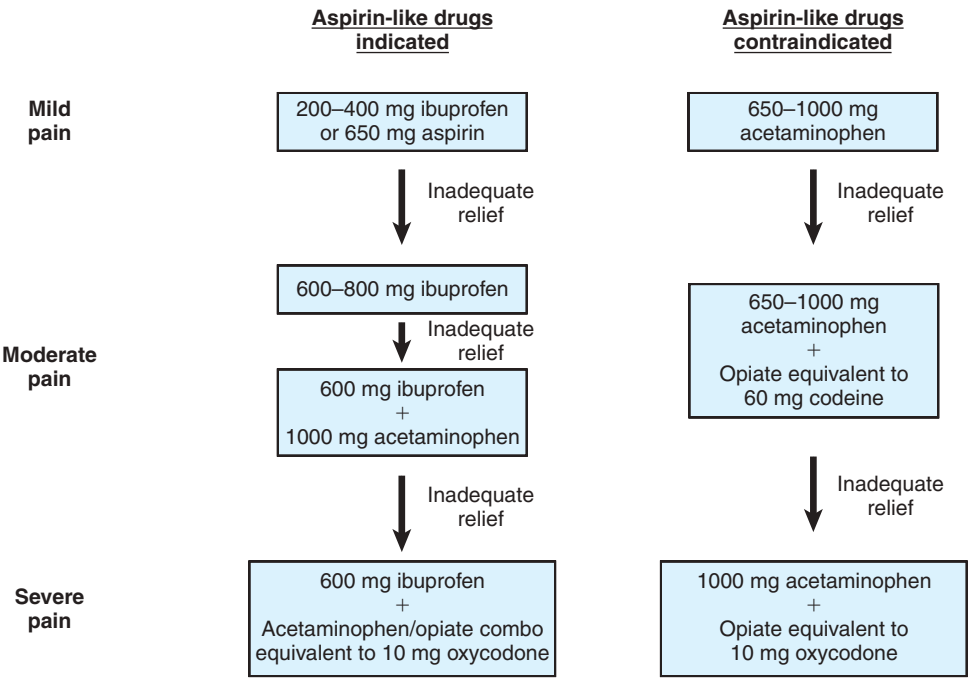


Fig. 6.21 A flexible analgesic strategy.

Administration of ibuprofen 600 mg with acetaminophen 1000 mg produced significant relief of posttreatment endodontic pain compared with ibuprofen alone or placebo (Fig. 6.22). However, in a recent study, the analgesic effect of a combination of ibuprofen 600 mg with acetaminophen 1000 mg did not differ from that of acetaminophen alone.[350] Studies have also shown that concurrent administration of an NSAID and an acetaminophen-opioid combination drug provided significantly greater analgesia than an NSAID alone.[37,323] The concurrent administration of acetaminophen and NSAID appears to be well tolerated, with no detectable increase in side effects or alterations in pharmacokinetics.[37,200,323,363]

In rare cases of severe pain, an NSAID may need to be administered with an opioid. Two general methods are used to combine NSAIDs and opioids to achieve the analgesic benefits of both. The first involves an alternating regimen of an NSAID followed by an acetaminophen-opioid combination.[13,66] Aspirin and opioid combinations are not used in this alternating schedule because of the potential for NSAID/aspirin interactions. The second method involves administration of a single drug consisting of an NSAID-opioid combination. For example, postoperative pain studies have shown that the combination of 200 mg of ibuprofen and 7.5 mg of hydrocodone (Vicoprofen) was about 80% more effective for analgesia than the ibuprofen alone, with about the same incidence of side effects.[353] Other opioids can be added to an NSAID for increased analgesia. A study on posttreatment endodontic pain demonstrated short-term benefits of the combination of flurbiprofen and

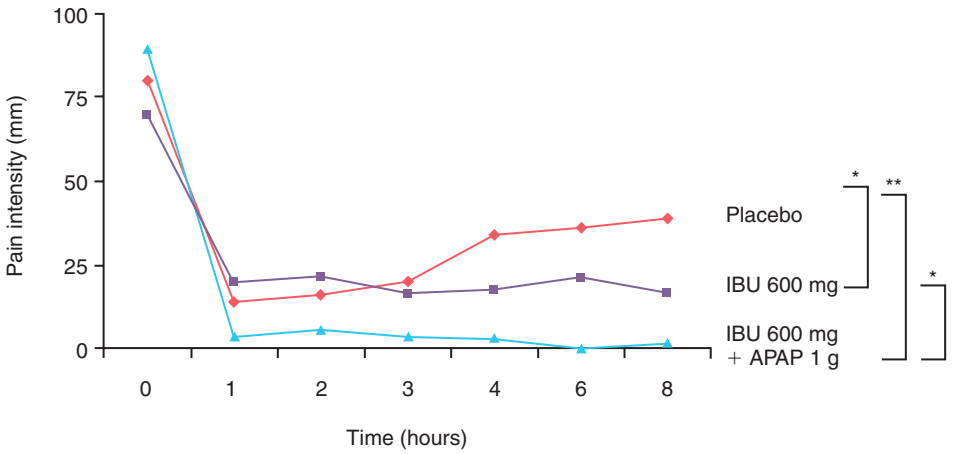


Fig. 6.22 Comparison of ibuprofen *(IBU)* 600 mg plus acetaminophen *(APAP)* 1000 mg with IBU alone or to placebo treatment in postendodontic pain patients. (From Menhinick KA, Gutmann JL, Regan JD, et al: The efficacy of pain control following nonsurgical root canal treatment using ibuprofen or a combination of ibuprofen and acetaminophen in a randomized, double-blind, placebo-controlled study, *Int Endod J* 37:531, 2004.)

tramadol.[86] Other NSAID and opioid combinations have also been evaluated.[83] However, the results of clinical trials on the use of NSAIDs alone and NSAIDs combined with acetaminophen (see Fig. 6.22) suggest that opioid combinations may be required only rarely.

Not all patients require concurrent use of NSAIDs with acetaminophen-opioid combinations or combinations of an NSAID and opioid. The basic premise of a flexible prescription plan is that the analgesic prescribed is matched to the patient's need. The major advantage of a flexible plan is that the clinician is prepared for those rare cases when additional pharmacotherapy is indicated to increase the efficacy of pain control. As mentioned, preoperative hyperalgesia may be an indicator for more comprehensive pharmacotherapy. When the clinician is considering the combination of various analgesics, she or he must make sure to use dosing regimens that do not exceed any of the drugs' maximal daily dosage.

Future Directions

COX-inhibiting nitric oxide (NO) donators (CINODs), a new class of NSAIDs, have a NO-donating moiety. These drugs, also known as NO-NSAIDs, were developed with the idea that the released NO would lead to improved vascular tone and mucosal blood flow, thus attenuating the adverse effects of NSAIDs on GI mucosa and blood pressure. Although none of the CINODS have yet been approved by the FDA, they may be available in the near future.

Current use of analgesics has been driven by clinical trials to arrive at doses that obtain the desired level of pain relief with acceptable side-effect profiles. After administration, drugs are absorbed and distributed to their site of action, where they interact with functional targets. They then undergo metabolism and eventual excretion. All of the steps along the way are influenced by a variety of environmental and genetic factors. The ability to predict how a patient's genome may affect the efficacy of a given analgesic drug is being discovered in the field of pain pharmacogenomics (for a review, see Rollason et al.[288]).

One example relevant to pain control in dentistry is the variable efficacy of codeine in a specific population. Many analgesic drugs are metabolized by the cytochrome P450 (CYP) family of hepatic enzymes, and the genes that encode for their biosynthesis have been identified. Codeine is a prodrug that is demethylated to produce morphine, which is responsible for its analgesic action. This demethylation is catalyzed by the enzyme cytochrome P402D6. It has been estimated that 6% to 7% of the Caucasian population has a nonfunctional CYP2D6 mutant allele, making them poor metabolizers of codeine to morphine.[79] These patients may be aware from past experience that codeine is ineffective in their systems and may request a different (usually more potent) form of narcotic. The clinician may suspect drug-seeking behavior when in fact there is a biochemical reason for such a request. Given the current state of clinical DNA analysis, it is not incomprehensible to envision a time when rapid (chairside) genomic evaluation will lead to specific recommendations for analgesic prescriptions.

Summary

The information and recommendations provided in this chapter were selected to aid the clinician in the management of acute endodontic pain. Clinical judgment must also take into account other sources of information—the patient history, concurrent medications, the nature of the pain, and the overall treatment plan—when the best means of alleviating patient pain are being determined. An effective approach to managing endodontic pain demands integration of the general principles of pain mechanisms and management with thorough, individualized clinical assessment.

References

1. Abbott A, et al: A prospective randomized trial on efficacy of antibiotic prophylaxis in asymptomatic teeth with pulpal necrosis and associated periapical pathosis, *Oral Surg* 66:722, 1988.
2. Aelig W, Laurence D, O'Neil R, et al: Cardiac effects of adrenaline and felypressin as vasoconstrictors in local anaesthesia for oral surgery under diazepam sedation, *Br J Anaesth* 42:174, 1970.
3. Aggarwal V, Jain A, Kabi D: Anesthetic efficacy of supplemental buccal and lingual infiltrations of articaine and lidocaine after an inferior alveolar nerve block in patients with irreversible pulpitis, *J Endod* 35:925, 2009.
4. Aggarwal V, Singla M, Kabi D: Comparative evaluation of anesthetic efficacy of Gow-Gates mandibular conduction anesthesia, Vazirani-Akinosi technique, buccal-plus-lingual infiltrations, and conventional inferior alveolar nerve anesthesia in patients with irreversible pulpitis, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 109:303, 2010.
5. Aggarwal V, Singla M, Miglani S, et al: Efficacy of articaine versus lidocaine administered as supplementary intraligamentary injection after a failed inferior alveolar nerve block: a randomized double-blind study, *J Endod* 45:1, 2019.
6. Aggarwal V, Singla M, Miglani S, et al: A prospective, randomized single-blind evaluation of effect of injection speed on anesthetic efficacy of inferior alveolar nerve block in patients with symptomatic irreversible pulpitis, *J Endod* 38:1578, 2012.
7. Aggarwal V, Singla M, Rizvi A, et al: Comparative evaluation of local infiltration of articaine, articaine plus ketorolac, and dexamethasone on anesthetic efficacy of inferior alveolar nerve block with lidocaine in patients with irreversible pulpitis, *J Endod* 37:445, 2011.
8. Agren E, Danielsson K: Conduction block analgesia in the mandible: a comparative investigation of the techniques of Fischer and Gow-Gates, *Swed Dent J* 5:91, 1981.
9. Akinosi J: A new approach to the mandibular nerve block, *Br J Oral Surg* 15:83, 1977.
10. Aldous J: Needle deflection: a factor in the administration of local anesthetics, *J Am Dent Assoc* 77:602, 1968.
11. Al-Sultan AF: Effectiveness of pH adjusted lidocaine versus commercial lidocaine for maxillary infiltration anesthesia, *Al-Rafidain Dent J* 4:34, 2004.
12. Al-Sultan AF, Fathie WK, Hamid RS: A clinical evaluation on the alkalization of local anesthetic solution in periapical surgery, *Al-Rafidain Dent J* 6:71, 2006.
13. American Association of Endodontists: *Post-endodontic pain control*, Chicago, 1995, The Association.
14. Anderson B: Paracetamol (acetaminophen): mechanisms of action, *Paediatr Anaesth* 18:915, 2008.
15. Arnold J, Salom I, Berger A: Comparison of gastrointestinal microbleeding associated with use of etodolac, ibuprofen, indomethacin, and naproxen in normal subjects, *Curr Ther Res* 37:730, 1985.
16. Asarch T, Allen K, Petersen B, et al: Efficacy of a computerized local anesthesia device in pediatric dentistry, *Pediatr Dent* 21:421, 1999.
17. Ashraf H, Kazem M, Dianat O, et al: Efficacy of articaine versus lidocaine in block and infiltration anesthesia administered in teeth with irreversible pulpitis: a prospective, randomized, double-blind study, *J Endod* 39:6, 2013.

18. Bacsik C, Swift J, Hargreaves K: Toxic systemic reactions of bupivacaine and etidocaine, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 79:18, 1995.
19. Balasco M, Drum M, Reader A, et al: Buffered lidocaine for incision and drainage: a prospective, randomized double-blind study, *J Endod* 39:1329, 2013.
20. Batinac T, Sotosek Tokmadzic V, Peharda V, et al: Adverse reactions and alleged allergy to local anesthetics: analysis of 331 patients, *J Dermatol* 40:522, 2013.
21. Batista da Silva C, Berto LA, Volpato MC, et al: Anesthetic efficacy of articaine and lidocaine for incisive/mental nerve block, *J Endod* 36:438, 2010.
22. Beaver W: Mild analgesics: a review of their clinical pharmacology, *Am J Med Sci* 251:576, 1966.
23. Benkwitz C, Garrison JC, Linden J, et al: Lidocaine enhances G alpha I protein function, Anesthesiology 99:1093, 2003.
24. Berberich G, Reader A, Drum M, et al: A prospective, randomized, double-blind comparison of the anesthetic efficacy of 2% lidocaine with 1 : 100,000 and 1 : 50,000 epinephrine and 3% mepivacaine in the intraoral, infraorbital nerve block, *J Endod* 35:1498, 2009.
25. Berlin J, Nusstein J, Reader A, et al: Efficacy of articaine and lidocaine in a primary intraligamentary injection administered with a computer-controlled local anesthetic delivery system, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 99:361, 2005.
26. Berns J, Sadove M: Mandibular block injection: a method of study using an injected radiopaque material, *J Am Dent Assoc* 65:736, 1962.
27. Bigby J, Reader A, Nusstein J, et al: Articaine for supplemental intraosseous anesthesia in patients with irreversible pulpitis, *J Endod* 32:1044, 2006.
28. Biocanin V, Brkovic B, Milicic B, et al: Efficacy and safety of intraseptal and periodontal ligament anesthesia achieved by computer-controlled articaine + epinephrine delivery: a dose-finding study, *Clin Oral Investig* 17:525, 2013.
29. Birchfield J, Rosenberg PA: Role of the anesthetic solution in intrapulpal anesthesia, *J Endod* 1:26, 1975.
30. Black JA, Liu S, Tanaka M, et al: Changes in the expression of tetrodotoxin-sensitive sodium channels within dorsal root ganglia neurons in inflammatory pain, *Pain* 108:237, 2004.
31. Bonar T, Nusstein J, Reader A, et al: Anesthetic efficacy of articaine and lidocaine in a primary intraseptal injection: a prospective, randomized double-blind study, *Anesth Prog* 64:203, 2017.
32. Bosco DA, Haas DA, Young ER, et al: An anaphylactoid reaction following local anesthesia: a case report, *Anesth Pain Control Dent* 2:87, 1993.
33. Bowles WR, Flores CM, Jackson DL, et al: Beta 2-adrenoceptor regulation of CGRP release from capsaicin-sensitive neurons, *J Dent Res* 82:308, 2003.
34. Bowles WH, Frysh H, Emmons R: Clinical evaluation of buffered local anesthetic, *Gen Dent* 43:182, 1995.
35. Brandt RG, Anderson PF, McDonald NJ, et al: The pulpal anesthetic efficacy of articaine versus lidocaine in dentistry: a meta-analysis, *J Am Dent Assoc* 142:493, 2011.
36. Brannstrom M, Lindskog S, Nordenvall K: Enamel hypoplasia in permanent teeth induced by periodontal ligament anesthesia of primary teeth, *J Am Dent Assoc* 109:735, 1984.
37. Breivik E, Barkvoll P, Skovlund E: Combining diclofenac with acetaminophen or acetaminophen-codeine after oral surgery: a randomized, double-blind, single oral dose study, *Clin Pharmacol Ther* 66:625, 1999.
38. Brkovic BM, Savic M, Andric M, et al: Intraseptal vs. periodontal ligament anaesthesia for maxillary tooth extraction: quality of local anaesthesia and haemodynamic response, *Clin Oral Investig* 14:675, 2010.
39. Broering R, Reader A, Drum M, et al: A prospective, randomized comparison of the anesthetic efficacy of the greater palatine and high tuberosity second division nerve blocks, *J Endod* 35:1337, 2009.
40. Brunetto PC, Ranali J, Ambrosano GMB, et al: Anesthetic efficacy of 3 volumes of lidocaine with epinephrine in maxillary infiltration anesthesia, *Anesth Prog* 55:29, 2008.
41. Bultema K, Fowler S, Drum M, et al: Pain reduction in untreated symptomatic irreversible pulpitis using liposomal bupivacaine (Exparel): a prospective, randomized, double-blind trial, *J Endod* 42:1707, 2016.
42. Bunczak-Reeh M, Hargreaves K: Effect of inflammation on delivery of drugs to dental pulp, *J Endod* 24:822, 1998.
43. Burns Y, Reader A, Nusstein J, et al: Anesthetic efficacy of the palatal-anterior superior alveolar injection, *J Am Dent Assoc* 135:1269, 2004.
44. Byers MR, Taylor PE, Khayat BG, et al: Effects of injury and inflammation on pulpal and periapical nerves, *J Endod* 16:78, 1990.
45. Byrne B: Drug interactions: a review and update, *Endod Topics* 4:9, 2004.
46. Cannell H, Kerwala C, Webster K, et al: Are intraligamentary injections intravascular? *Br Dent J* 175:281, 1993.
47. Cairns BE, Dong XD, Wong H, et al: Intramuscular ketorolac inhibits activation of rat peripheral NMDA receptors, *J Neurophysiol* 107:3308, 2012.
48. Cepeda MS, Tzortzopoulou A, Thackrey M, et al: Adjusting the pH of lidocaine for reducing pain on injection (review), *Cochrane Database Syst Rev* 8:12, 2010.
49. Certosimo A, Archer R: A clinical evaluation of the electric pulp tester as an indicator of local anesthesia, *Oper Dent* 21:25, 1996.
50. Chamberlain T, Davis R, Murchison D, et al: Systemic effects of an intraosseous injection of 2% lidocaine with 1:,Äâ100,000 epinephrine, *Gen Dent* 48:299, 2000.
51. Chance K, Lin L, Shovlin F, et al: Clinical trial of intracanal corticosteroid in root canal therapy, *J Endod* 13:466, 1987.
52. Chandrasekharan NV, Dai H, Roos KL, et al: COX-3, a cyclooxygenase-1 variant inhibited by acetaminophen and other analgesic/antipyretic drugs: cloning, structure, and expression [see comment], *Proc Natl Acad Sci U S A* 99:13926, 2002.
53. Chaney MA, Kerby R, Reader A, et al: An evaluation of lidocaine hydrocarbonate compared with lidocaine hydrochloride for inferior alveolar nerve block, *Anesth Prog* 38:212, 1991.
54. Childers M, Reader A, Nist R, et al: Anesthetic efficacy of the periodontal ligament injection after an inferior alveolar nerve block, *J Endod* 22:317, 1996.
55. Chiu CY, Lin TY, Hsia SH, et al: Systemic anaphylaxis following local lidocaine administration during a dental procedure, *Pediatr Emerg Care* 20:178, 2004.
56. Ciancio SG, Hutcheson MC, Ayoub F, et al: Safety and efficacy of a novel nasal spray for maxillary dental anesthesia, *J Dent Res* 92:43S, 2013.
57. Claffey E, Reader A, Nusstein J, et al: Anesthetic efficacy of articaine for inferior alveolar nerve blocks in patients with irreversible pulpitis, *J Endod* 30:568, 2004.
58. Clark S, Reader A, Beck M, et al: Anesthetic efficacy of the mylohyoid nerve block and combination inferior alveolar nerve block/mylohyoid nerve block, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 87:557, 1999.
59. Clause DW, Zach GW: Reaction to diphenhydramine hydrochloride (Benadryl) used as a local anesthetic, *Gen Dent* 37:426, 1989.
60. Click V, Drum M, Reader A, et al: Evaluation of the Gow-Gates and Vazirani-Akinosi techniques in patients with symptomatic irreversible pulpitis: a prospective randomized study, *J Endod* 41:16, 2015.
61. Coggins R, Reader A, Nist R, et al: Anesthetic efficacy of the intraosseous injection in maxillary and mandibular teeth, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 81:634, 1996.
62. Cohen H, Cha B, Spangberg L: Endodontic anesthesia in mandibular molars: a clinical study, *J Endod* 19:370, 1993.
63. *Drug facts and comparisons*, St. Louis, 2000, Facts and Comparisons, Inc.
64. Cooley R, Robison S: Comparative evaluation of the 30-gauge dental needle, *Oral Surg Oral Med Oral Pathol* 48:400, 1979.
65. Cooper S: New peripherally acting oral analgesics, *Annu Rev Pharmacol Toxicol* 23:617, 1983.
66. Cooper S: Treating acute dental pain, *Postgrad Dent* 2:7, 1995.
67. Cooper S, Berrie R, Cohn P: The analgesic efficacy of ketoprofen compared to ibuprofen and placebo, *Adv Ther* 5:43, 1988.
68. Cooper SA: The relative efficacy of ibuprofen in dental pain, *Compend Contin Educ Dent* 7:580, 1986.
69. Corbett IP, Kanaa MD, Whitworth JM, et al: Articaine infiltration for anesthesia of mandibular first molars, *J Endod* 34:514, 2008.
70. Cromley N, Adams D: The effect of intraligamentary injections on diseased periodontiums in dogs, *Gen Dent* 39:33, 1991.
71. Crout R, Koraido G, Moore P: A clinical trial of long-acting local anesthetics for periodontal surgery, *Anesth Prog* 37:194, 1990.
72. D'Souza J, Walton R, Peterson L: Periodontal ligament injection: an evaluation of extent of anesthesia and postinjection discomfort, *J Am Dent Assoc* 114:341, 1987.

73. Dagher FB, Yared GM, Machtou P: An evaluation of 2% lidocaine with different concentrations of epinephrine for inferior alveolar nerve block, *J Endod* 23:178, 1997.
74. Danielsson K, Evers H, Nordenram A: Long-acting local anesthetics in oral surgery: an experimental evaluation of bupivacaine and etidocaine for oral infiltration anesthesia, *Anesth Prog* 32:65, 1985.
75. Davidson M: Bevel-oriented mandibular injections: needle deflection can be beneficial, *Gen Dent* 37:410, 1989.
76. Davis W, Oakley J, Smith E: Comparison of the effectiveness of etidocaine and lidocaine as local anesthetic agents during oral surgery, *Anesth Prog* 31:159, 1984.
77. DeJong R: Neural blockade by local anesthetics, *J Am Dent Assoc* 238:1383, 1997.
78. Dernedde M, Furlan D, Verbesselt R, et al: Grand mal convulsion after an accidental intravenous injection of ropivacaine, *Anesth Analg* 98:521, 2004.
79. Diogenes A, Akopian AN, Hargreaves KM: NGF upregulates TRPA1: implications for orofacial pain, *J Dent Res* 86:550, 2007.
80. Dionne R: Suppression of dental pain by the preoperative administration of flurbiprofen, *Am J Med Sci* 80:41, 1986.
81. Dionne R: COX-2 inhibitors: better than ibuprofen for dental pain? *Compendium* 20:518, 1999.
82. Dionne RA: Additive analgesic effects of oxycodone and ibuprofen in the oral surgery model, *J Oral Maxillofac Surg* 57:673, 1999.
83. Dionne RA, Berthold C: Therapeutic uses of non-steroidal anti-inflammatory drugs in dentistry, *Crit Rev Oral Biol Med* 12:315, 2000.
84. Dionne RA, Lepinski AM, Gordon SM, et al: Analgesic effects of peripherally administered opioids in clinical models of acute and chronic inflammation, *Clin Pharmacol Ther* 70:66, 2001.
85. Dong XD, Svensson P, Cairns BE: The analgesic action of topical diclofenac may be mediated through peripheral NMDA receptor antagonism, *Pain* 15:36, 2009.
86. Doroshak A, Bowles W, Hargreaves K: Evaluation of the combination of flurbiprofen and tramadol for management of endodontic pain, *J Endod* 25:660, 1999.
87. Dou L, Luo J, Yang D: Anaesthetic efficacy of supplemental lingual infiltration of mandibular molars after inferior alveolar nerve block plus buccal infiltration in patients with irreversible pulpitis, *Int Endod J* 120:42, 2012.
88. Dreven LJ, Reader A, Beck M, et al: An evaluation of an electric pulp tester as a measure of analgesia in human vital teeth, *J Endod* 13:233, 1987.
89. Dreyer WP, van Heerden JD, de V Joubert JJ: The route of periodontal ligament injection of local anesthetic solution, *J Endod* 9:471, 1983.
90. Dunbar D, Reader A, Nist R, et al: Anesthetic efficacy of the intraosseous injection after an inferior alveolar nerve block, *J Endod* 22:481, 1996.
91. Dunsky JL, Moore PA: Long-acting local anesthetics: a comparison of bupivacaine and etidocaine in endodontics, *J Endod* 10:457, 1984.
92. Edwards R, Head T: A clinical trial of intraligamentary anesthesia, *J Dent Res* 68:1210, 1989.
93. Ehrmann EH, Messer HH, Adams GG: The relationship of intracanal medicaments to postoperative pain in endodontics, *Int Endod J* 36:868, 2003.
94. Evans G, Nusstein J, Drum M, et al: A prospective, randomized, double-blind comparison of articaine and lidocaine for maxillary infiltrations, *J Endod* 34:389, 2008.
95. Fan S, Chen WL, Pan CB, et al: Anesthetic efficacy of inferior alveolar nerve block plus buccal infiltration or periodontal ligament injections with articaine in patients with irreversible pulpitis, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 108:89, 2009.
96. Farhad A, Razavian H, Shafiee M: Effect of intraosseous injection versus inferior alveolar nerve block as primary pulpal anaesthesia of mandibular posterior teeth with symptomatic irreversible pulpitis: a prospective randomized clinical trial, *Acta Odontol Scand* 76:442, 2018.
97. Fehrenbacher J, Sun XX, Locke E, et al: Capsaicin-evoked iCGRP release from human dental pulp: a model system for the study of peripheral neuropeptide secretion in normal healthy tissue, *Pain* 144:253, 2009.
98. Fernandez C, Reader A, Beck M, et al: A prospective, randomized, double-blind comparison of bupivacaine and lidocaine for inferior alveolar nerve blocks, *J Endod* 31:499, 2005.
99. Fernieini EM, Bennett JD, Silverman DG, et al: Hemodynamic assessment of local anesthetic administration by laser Doppler flowmetry, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 91:526, 2001.
100. Finder R, Moore PA: Adverse drug reactions to local anesthesia, *Dent Clin North Am* 46:747, 2002.
101. Forloine A, Drum M, Reader A, et al: A prospective, randomized, double-blind comparison of the anesthetic efficacy of two percent lidocaine with 1:100,000 epinephrine and three percent mepivacaine in the maxillary high tuberosity second division nerve block, *J Endod* 36:1770, 2010.
102. Fouad A: Are antibiotics effective for endodontic pain? An evidence-based review, *Endod Topics* 3:52, 2002.
103. Fouad A, Rivera E, Walton R: Penicillin as a supplement in resolving the localized acute apical abscess, *Oral Surg Oral Med Oral Pathol* 81:590, 1996.
104. Fowler S, Drum M, Reader A, et al: Anesthetic success of an inferior alveolar nerve block and supplemental articaine buccal infiltration for molars and premolars in patients with symptomatic irreversible pulpitis, *J Endod* 42:390, 2016.
105. Fowler S, Reader A: Is the volume of 3.6 mL better than 1.8 mL for inferior alveolar nerve blocks in patients with symptomatic irreversible pulpitis? *J Endod* 39:970, 2013.
106. Fowler S, Reader A, Beck M: Incidence of missed inferior alveolar nerve blocks in vital asymptomatic subjects and in patients with symptomatic irreversible pulpitis, *J Endod* 41:637, 2015.
107. Friedman M, Hochman M: A 21st century computerized injection system for local pain control, *Compendium* 18:995, 1997.
108. Friedman M, Hochman M: The AMSA injection: a new concept for local anesthesia of maxillary teeth using a computer-controlled injection system, *Quintessence Int* 29:297, 1998.
109. Friedman M, Hochman M: P-ASA block injection: a new palatal technique to anesthetize maxillary anterior teeth, *J Esthet Dent* 11:63, 1999.
110. Friedman M, Hochman M: Using AMSA and P-ASA nerve blocks for esthetic restorative dentistry, *Gen Dent* 5:506, 2001.
111. Frommer J, Mele F, Monroe C: The possible role of the mylohyoid nerve in mandibular posterior tooth sensation, *J Am Dent Assoc* 85:113, 1972.
112. Froum SJ, Tarnow D, Caiazzo A, et al: Histologic response to intra-ligament injections using a computerized local anesthetic delivery system: a pilot study in mini-swine, *J Periodontol* 71:1453, 2000.
113. Fuhs QM, Walker WA III, Gough RW, et al: The periodontal ligament injection: histological effects on the periodontium in dogs, *J Endod* 9:411, 1983.
114. Gaffen AS, Haas DA: Retrospective review of voluntary reports of nonsurgical paresthesia in dentistry, *J Can Dent Assoc* 75:579, 2009.
115. Gage T, Pickett F: *Mosby's dental drug reference*, ed 4, St. Louis, 2000, Mosby.
116. Galbreath J: Tracing the course of the mandibular block injection, *Oral Surg Oral Med Oral Pathol* 30:571, 1970.
117. Galili D, Kaufman E, Garfunkel AA, et al: Intraligamental anesthesia: a histological study, *Int J Oral Surg* 13:511, 1984.
118. Gallatin E, Stabile P, Reader A, et al: Anesthetic efficacy and heart rate effects of the intraosseous injection of 3% mepivacaine after an inferior alveolar nerve block, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 89:83, 2000.
119. Gallatin J, Nusstein J, Reader A, et al: A comparison of injection pain and postoperative pain of two intraosseous anesthetic techniques, *Anesth Prog* 50:111, 2003.
120. Gallatin J, Reader A, Nusstein J, et al: A comparison of two intraosseous anesthetic techniques in mandibular posterior teeth, *J Am Dent Assoc* 134:1476, 2003.
121. Garisto GA, Gaffen AS, Lawrence HP, et al: Occurrence of paresthesia after dental local anesthetic administration in the United States, *J Am Dent Assoc* 141:836, 2010.
122. Gear RW, Miaskowski C, Gordon NC, et al: Kappa-opioids produce significantly greater analgesia in women than in men, *Nat Med* 2:1248, 1996.
123. Geborek P, Mansson B, Wollheim FA, et al: Intraarticular corticosteroid injection into rheumatoid arthritis knees improves extensor muscles strength, *Rheumatol Int* 9:265, 1990.
124. Giannakopoulos H, Levin LM, Chou JC, et al: The cardiovascular effects and pharmacokinetics of intranasal tetracaine plus oxymetazoline: preliminary findings, *J Am Dent Assoc* 143: 872, 2012.
125. Glassman G, Krasner P, Morse DR, et al: A prospective randomized double-blind trial on efficacy of dexamethasone for endodontic interappointment pain in teeth with asymptomatic inflamed pulps, *Oral Surg Oral Med Oral Pathol* 67:96, 1989.

126. Glenn B, Drum M, Reader A, et al: Does liposomal bupivacaine (Exparel) significantly reduce postoperative pain/numbness in symptomatic teeth with a diagnosis of necrosis? A prospective, randomized, double-blind trial, *J Endod* 42:1301, 2016.
127. Gold M, Reichling D, Shuster M, et al: Hyperalgesic agents increase a tetrodotoxin-resistant Na^+-current in nociceptors, *Proc Natl Acad Sci U S A* 93:1108, 1996.
128. Goldberg S, Reader A, Drum M, et al: A comparison of the anesthetic efficacy of the conventional inferior alveolar, Gow-Gates and Vazirani-Akinosi techniques, *J Endod* 34:1306, 2008.
129. Goldstein DS, Dionne R, Sweet J, et al: Circulatory, plasma catecholamine, cortisol, lipid, and psychological responses to a real-life stress (third molar extractions): effects of diazepam sedation and of inclusion of epinephrine with the local anesthetic, *Psychosom Med* 44:259, 1982.
130. Gordon SM: Blockade of peripheral neuronal barrage reduces postoperative pain, *Pain* 306:264, 1997.
131. Gordon SM, Brahim JS, Dubner R, et al: Attenuation of pain in a randomized trial by suppression of peripheral nociceptive activity in the immediate postoperative period, *Anesth Analg* 95:1351, 2002.
132. Gow-Gates G: Mandibular conduction anesthesia: a new technique using extra-oral landmarks, *Oral Surg Oral Med Oral Pathol* 36:321, 1973.
133. Gray R, Lomax A, Rood J: Periodontal ligament injection: with or without a vasoconstrictor? *Br Dent J* 162:263, 1987.
134. Gross R, McCartney M, Reader A, et al: A prospective, randomized, double-blind comparison of bupivacaine and lidocaine for maxillary infiltrations, *J Endod* 33:1021, 2007.
135. Guglielmo A, Reader A, Nist R, et al: Anesthetic efficacy and heart rate effects of the supplemental intraosseous injection of 2% mepivacaine with 1:20,000 levonordefrin, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 87:284, 1999.
136. Haas DA, Lennon D: A 21 year retrospective study of reports of paresthesia following local anesthetic administration, *J Can Dent Assoc* 61:319, 1995.
137. Haas DA, Pynn BR, Sands TD: Drug use for the pregnant or lactating patient, *Gen Dent* 48:54, 2000.
138. Haase A, Reader A, Nusstein J, et al: Comparing anesthetic efficacy of articaine versus lidocaine as a supplemental buccal infiltration of the mandibular first molar after an inferior alveolar nerve block, *J Am Dent Assoc* 139:1228, 2008.
139. Hanna MN, Elhassan A, Veloso PM, et al: Efficacy of bicarbonate in decreasing pain on intradermal injection of local anesthetics: a meta-analysis, *Reg Anesth Pain Med* 34:122, 2009.
140. Hannan L, Reader A, Nist R, et al: The use of ultrasound for guiding needle placement for inferior alveolar nerve blocks, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 87:658, 1999.
141. Harboe T, Guttormsen AB, Aarebrot S, et al: Suspected allergy to local anaesthetics: follow-up in 135 cases, *Acta Anaesthesiol Scand* 54:536, 2010.
142. Hargreaves K: Neurochemical factors in injury and inflammation in orofacial tissues. In Lund JP, Lavigne GJ, Dubner R, et al., editors: *Orofacial pain: basic sciences to clinical management*, Chicago, 2000, Quintessence.
143. Hargreaves K, Joris J: The peripheral analgesic effects of opioids, *APS J* 2:51, 1993.
144. Hargreaves K, Troullos E, Dionne R: Pharmacologic rationale for the treatment of acute pain, *Dent Clin North Am* 31:675, 1987.
145. Hargreaves KM, Jackson DL, Bowles WR: Adrenergic regulation of capsaicin-sensitive neurons in dental pulp, *J Endod* 29:397, 2003.
146. Hargreaves KM, Keiser K: Local anesthetic failure in endodontics: mechanisms and management, *Endod Topics* 1:26, 2003.
147. Hargreaves KM, Keiser K: New advances in the management of endodontic pain emergencies, *J Calif Dent Assoc* 32:469, 2004.
148. Harreld TK, Fowler S, Drum M, et al: Efficacy of a buffered 4% lidocaine formulation for incision and drainage: a prospective, randomized, double-blind study, *J Endod* 41:1583, 2015.
149. Hasse AL, Heng MK, Garrett NR: Blood pressure and electrocardiographic response to dental treatment with use of local anesthesia, *J Am Dent Assoc* 113:639, 1986.
150. Hasselgren G, Reit C: Emergency pulpotomy: pain relieving effect with and without the use of sedative dressings, *J Endod* 15:254, 1989.
151. Hersh EV, Lally ET, Moore PA: Update on cyclooxygenase inhibitors: Has a third COX isoform entered the fray? *Curr Med Res Opin* 21:1217, 2005.
152. Hersh EV, Pinto A, Saraghi M, et al: Double-masked, randomized, placebo-controlled study to evaluate the efficacy and tolerability of intranasal K305 (3% tetracaine plus 0.05% oxymetazoline) in anesthetizing maxillary teeth, *J Am Dent Assoc* 147:278, 2016.
153. Hersh EV, Saraghi M, Moore PA: Intranasal tetracaine and oxymetazoline: a newly approved drug formulation that provides maxillary dental anesthesia without needles, *Curr Med Res Opin* 3:1, 2016.
154. Hinkley SA, Reader A, Beck M, et al: An evaluation of 4% prilocaine with 1:200,000 epinephrine and 2% mepivacaine with 1:20,000 levonordefrin compared with 2% lidocaine with 1:100,000 epinephrine for inferior alveolar nerve block, *Anesth Prog* 38:84, 1991.
155. Hobeich P, Simon S, Schneiderman E, et al: A prospective, randomized, double-blind comparison of the injection pain and anesthetic onset of 2% lidocaine with 1:100,000 epinephrine buffered with 5% and 10% sodium bicarbonate in maxillary infiltrations, *J Endod* 39:597, 2013.
156. Hochman M, Friedman M: In vitro study of needle deflection: a linear insertion technique versus a bidirectional rotation insertion technique, *Quintessence Int* 31:33, 2000.
157. Hollmann MW, Herroeder S, Kurz KS, et al: Time-dependent inhibition of G protein-coupled receptor signaling by local anesthetics, *Anesthesiology* 100:852, 2004.
158. Holstein A, Hargreaves KM, Niederman R: Evaluation of NSAIDs for treating post-endodontic pain, *Endod Topics* 3:3, 2002.
159. Hull T, Rothwell B: Intraosseous anesthesia comparing lidocaine and etidocaine (abstract), *J Dent Res* 77:197, 1998.
160. Ingle J, Bakland L: *Endodontics* (vol 6), Hamilton, Ontario, 2008, Decker.
161. Jaber A, Whitworth JM, Corbett IP, et al: The efficacy of infiltration anaesthesia for adult mandibular incisors: a randomized double-blind cross-over trial comparing articaine and lidocaine buccal and buccal plus lingual infiltrations, *Br Dent J* 209:E16, 2010.
162. Jackson D, Moore P, Hargreaves K: Preoperative nonsteroidal anti-inflammatory medication for the prevention of postoperative dental pain, *J Am Dent Assoc* 119:641, 1989.
163. Jage J: Circulatory effects of vasoconstrictors combined with local anesthetics, *Anesth Pain Control Dent* 2:81, 1993.
164. Jalalzadeh SM, Mamavi A, Shahriari S, et al: Effect of pretreatment prednisolone on postendodontic pain: a double-blind parallel-randomized clinical trial, *J Endod* 36:978, 2010.
165. Jastak J, Yagiela J: *Local anesthesia of the oral cavity*, New York, 1995, Elsevier Health Science.
166. Jebeles JA, Reilly JS, Gutierrez JF, et al: Tonsillectomy and adenoidectomy pain reduction by local bupivacaine infiltration in children, *Int J Pediatr Otorhinolaryngol* 25:149, 1993.
167. Jensen J, Nusstein J, Drum M, et al: Anesthetic efficacy of a repeated intraosseous injection following a primary intraosseous injection, *J Endod* 34:126, 2008.
168. Jeske A, Boschart B: Deflection of conventional versus nondeflecting dental needles in vitro, *Anesth Prog* 32:62, 1985.
169. Johnson G, Hlava G, Kalkwarf K: A comparison of periodontal intraligamental anesthesia using etidocaine HCL and lidocaine HCL, *Anesth Prog* 32:202, 1985.
170. Jones VR, Rivera EM, Walton RE: Comparison of carbon dioxide versus refrigerant spray to determine pulpal responsiveness, *J Endod* 28:531, 2002.
171. Jung IY, Kim JH, Kim ES, et al: An evaluation of buccal infiltrations and inferior alveolar nerve blocks in pulpal anesthesia for mandibular first molars, *J Endod* 34:11, 2008.
172. Kanaa MD, Meechan JG, Corbett IP, et al: Speed of injection influences efficacy of inferior alveolar nerve blocks: a double-blind randomized controlled trial in volunteers, *J Endod* 32:919, 2006.
173. Kanaa MD, Whitworth JM, Corbett IP, et al: Articaine and lidocaine mandibular buccal infiltration anesthesia; a prospective randomized double-blind cross-over study, *J Endod* 32:296, 2006.
174. Kanaa MD, Whitworth JM, Corbett IP, et al: Articaine buccal infiltration enhances the effectiveness of lidocaine inferior alveolar nerve block, *Int Endod J* 42:238, 2009.
175. Kanaa MD, Whitworth JM, Meechan JG: A prospective trial of different supplementary local anesthetic techniques after failure of inferior alveolar nerve block in patients with irreversible pulpitis in mandibular teeth, *J Endod* 38:421, 2012.
176. Karkut B, Reader A, Drum M, et al: A comparison of the local anesthetic efficacy of the extraoral versus the intraoral infraorbital nerve block, *J Am Dent Assoc* 141:185, 2010.

177. Kashyap VM, Desai R, Reddy PB, et al: Effect of alkalinisation of lignocaine for intraoral nerve block on pain during injection, and speed of onset of anaesthesia, *Br J Oral Maxillofac Surg* 49:e72, 2011.
178. Katz S, Drum M, Reader A, et al: A prospective, randomized, double-blind comparison of 2% lidocaine with 1:100,000 epinephrine, 4% prilocaine with 1:200,000 epinephrine and 4% prilocaine for maxillary infiltrations, *Anesth Prog* 57:45, 2010.
179. Kaufman E, Solomon V, Rozen L, et al: Pulpal efficacy of four lidocaine solutions injected with an intraligamentary syringe, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 78:17, 1994.
180. Kaufman E, Weinstein P, Milgrom P: Difficulties in achieving local anesthesia, *J Am Dent Assoc* 108:205, 1984.
181. Kaufman E, Heling I, Rotstein I, et al: Intraligamentary injection of slow-release methylprednisolone for the prevention of pain after endodontic treatment, *Oral Surg Oral Med Oral Pathol* 77:651, 1994.
182. Kearney PM, Baigent C, Godwin J, et al: Do selective cyclo-oxygenase-2 inhibitors and traditional non-steroidal anti-inflammatory drugs increase the risk of atherothrombosis? Meta-analysis of randomised trials [see comment], *Br Med J* 332:1302, 2006.
183. Keenan JV, Farman A, Fedorowicz Z, et al: A Cochrane systematic review finds no evidence to support the use of antibiotics for pain relief in irreversible pulpitis, *J Endod* 32:87, 2006.
184. Keiser K: Strategies for managing the endodontic pain patient, *Tex Dent J* 120:250, 2003.
185. Keiser K, Hargreaves K: Building effective strategies for the management of endodontic pain, *Endod Topics* 3:93, 2002.
186. Keiser K, Hargreaves KM: Strategies for managing the endodontic pain patient, *J Tenn Dent Assoc* 83:24, 2003.
187. Kennedy M, Reader A, Beck M, et al: Anesthetic efficacy of ropivacaine in maxillary anterior infiltration, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 91:406, 2001.
188. Kerckhove N, Mallet C, François A, et al: Cav3.2 calcium channels: the key protagonist in the supraspinal effect of paracetamol, *Pain* 155:764, 2014.
189. Khan AA, Dionne RA: The COX-2 inhibitors: new analgesic and anti-inflammatory drugs, *Dent Clin North Am* 46:679, 2002.
190. Kim S: Ligamental injection: a physiological explanation of its efficacy, *J Endod* 12:486, 1986.
191. Kindler CH, Paul M, Zou H, et al: Amide local anesthetics potently inhibit the human tandem pore domain background K+ channel TASK-2 (KCNK5), *J Pharmacol Exp Ther* 306:84, 2003.
192. Kis B, Snipes A, Bari F, et al: Regional distribution of cyclooxygenase-3 mRNA in the rat central nervous system, *Brain Res Mol Brain Res* 126:78, 2004.
193. Klein SM, Pierce T, Rubin Y, et al: Successful resuscitation after ropivacaine-induced ventricular fibrillation, *Anesth Analg* 97:901, 2003 [erratum appears in *Anesth Analg* 98:200, 2004].
194. Knoll-Kohler E, Frie A, Becker J, et al: Changes in plasma epinephrine concentration after dental infiltration anesthesia with different doses of epinephrine, *J Dent Res* 68:1098, 1989.
195. Knoll-Kohler E, Knoller M, Brandt K, et al: Cardiohemodynamic and serum catecholamine response to surgical removal of impacted mandibular third molars under local anesthesia: a randomized double-blind parallel group and crossover study, *J Oral Maxillofac Surg* 49:957, 1991.
196. Krasner P, Jackson E: Management of posttreatment endodontic pain with oral dexamethasone: a double-blind study, *Oral Surg Oral Med Oral Pathol* 62:187, 1986.
197. Kung J, McDonagh M, Sedgley CM: Does articaine provide an advantage over lidocaine in patients with symptomatic irreversible pulpitis? A systematic review and meta-analysis, *J Endod* 41:1784, 2015.
198. Lai J, Porreca J, Hunter J, et al: Voltage-gated sodium channels and hyperalgesia, *Annu Rev Pharmacol* 44:371, 2004.
199. Laine L, Bombardier C, Hawkey CJ, et al: Stratifying the risk of NSAID-related upper gastrointestinal clinical events: results of a double-blind outcomes study in patients with rheumatoid arthritis, *Gastroenterology* 123:1006, 2002.
200. Lanza FL, Royer GL, Nelson RS, et al: Effect of acetaminophen on human gastric mucosal injury caused by ibuprofen, *Gut* 27:440, 1986.
201. Larson AM, Polson J, Fontana RJ, et al: Acute liver failure study G: Acetaminophen-induced acute liver failure: results of a United States multicenter, prospective study [see comment], *Hepatology* 42:1364, 2005.
202. Lawaty I, Drum M, Reader A, et al: A prospective, randomized, double-blind comparison of 2% mepivacaine with 1:20,000 levonordefrin versus 2% lidocaine with 1:100,000 epinephrine for maxillary infiltrations, *Anesth Prog* 57:139, 2010.
203. Lee S, Reader A, Nusstein J, et al: Anesthetic efficacy of the anterior middle superior alveolar (AMSA) injection, *Anesth Prog* 51:80, 2004.
204. Liesinger A, Marshall F, Marshall J: Effect of variable doses of dexamethasone on posttreatment endodontic pain, *J Endod* 19: 35, 1993.
205. Lin L, Lapeyrolerie M, Skribner J, et al: Periodontal ligament injection: effects on pulp tissue, *J Endod* 11:529, 1985.
206. Lin S, Levin L, Emodi O, et al: Etodolac versus dexamethasone effect in reduction of postoperative symptoms following surgical endodontic treatment: a double-blind study, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 101:814, 2006.
207. Linden E, Abrams H, Matheny J, et al: A comparison of postoperative pain experience following periodontal surgery using two local anesthetic agents, *J Periodontol* 57:637, 1986.
208. List G, Meinster F Jr, Nery EB, et al: Gingival crevicular fluid response to various solutions using the intraligamentary injection, *Quintessence Int* 19:559, 1988.
209. Littner MM, Tamse A, Kaffe I: A new technique of selective anesthesia for diagnosing acute pulpitis in the mandible, *J Endod* 9:116, 1983.
210. Loetscher C, Melton D, Walton R: Injection regimen for anesthesia of the maxillary first molar, *J Am Dent Assoc* 117:337, 1988.
211. Malamed S: The Gow-Gates mandibular block: evaluation after 4,275 cases, *Oral Surg Oral Med Oral Pathol* 51:463, 1981.
212. Malamed S: The periodontal ligament (PDL) injection: an alternative to inferior alveolar nerve block, *Oral Surg Oral Med Oral Pathol* 53:117, 1982.
213. Malamed S: *Handbook of local anesthesia*, ed 6, St. Louis, 2013, Mosby/Elsevier.
214. Malamed SF, Gagnon S, Leblanc D: Articaine hydrochloride: a study of the safety of a new amide local anesthetic, *J Am Dent Assoc* 132:177, 2001.
215. Malav K, Singariya G, Mohammed S, et al: Comparison of 0.5% ropivacaine and 0.5% levobupivacaine for sciatic nerve block using Labat approach in foot and ankle surgery, *Turk J Anaesthesiol Reanim* 46:15, 2018.
216. Malmberg A, Yaksh T: Antinociceptive actions of spinal nonsteroidal anti-inflammatory agents on the formalin test in rats, *J Pharmacol Exp Ther* 263:136, 1992.
217. Marshall G: Consideration of steroids for endodontic pain, *Endod Topics* 3:41, 2002.
218. Marshall J, Walton R: The effect of intramuscular injection of steroid on posttreatment endodontic pain, *J Endod* 10:584, 1984.
219. Martinez G, Benito P, Fernandez C, et al: A comparative study of direct mandibular nerve block and the Akinosi technique, *Med Oral* 8:143, 2003.
220. Mason R, Drum M, Reader A, et al: A prospective, randomized, double-blind comparison of 2% lidocaine with 1:100,000 and 1:50,000 epinephrine and 3% mepivacaine for maxillary infiltrations, *J Endod* 35:1173, 2009.
221. Matthews R, Drum M, Reader A, et al: Articaine for supplemental, buccal mandibular infiltration anesthesia in patients with irreversible pulpitis when the inferior alveolar nerve block fails, *J Endod* 35:343, 2009.
222. McDougal RA, Delano EO, Caplan D, et al: Success of an alternative for interim management of irreversible pulpitis, *J Am Dent Assoc* 135:1707, 2004.
223. McEntire M, Nusstein J, Drum M, et al: Anesthetic efficacy of 4% articaine with 1:100,000 epinephrine versus 4% articaine with 1:200,000 epinephrine as a primary buccal infiltration in the mandibular first molar, *J Endod* 37:450, 2011.
224. McGettigan P, Henry D: Cardiovascular risk and inhibition of cyclooxygenase: a systematic review of the observational studies of selective and nonselective inhibitors of cyclooxygenase 2 [see comment], *JAMA* 296:1633, 2006.
225. McLean C, Reader A, Beck M, et al: An evaluation of 4% prilocaine and 3% mepivacaine compared with 2% lidocaine (1:100,000 epinephrine) for inferior alveolar nerve block, *J Endod* 19:146, 1993.
226. Meechan J: A comparison of ropivacaine and lidocaine with epinephrine for intraligamentary anesthesia, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 93:469, 2002.

227. Meechan J, Ledvinka J: Pulpal anesthesia for mandibular central incisor teeth: a comparison of infiltration and intraligamentary injections, *Int Endod J* 35:629, 2002.
228. Meechan JG, Rawlins MD: The effects of two different dental local anesthetic solutions on plasma potassium levels during third molar surgery, *Oral Surg Oral Med Oral Pathol* 66:650, 1988.
229. Mehrvarzfar P, Shababi B, Sayyad R, et al: Effect of supraperiosteal injection of dexamethasone on postoperative pain, *Aust Endod J* 34:25, 2008.
230. Menhinick K, Gutmann J, Regan J, et al: The efficacy of pain control following nonsurgical root canal treatment using ibuprofen or a combination of ibuprofen and acetaminophen in a randomized, double-blind, placebo-controlled study, *Int Endod J* 37:531, 2003.
231. Mikesell A, Drum M, Reader A, et al: Anesthetic efficacy of 1.8 mL and 3.6 mL of 2% lidocaine with 1:100,000 epinephrine for maxillary infiltrations, *J Endod* 34:121, 2008.
232. Modaresi J, Dianat O, Soluti A: Effect of pulp inflammation on nerve impulse quality with or without anesthesia, *J Endod* 34:438, 2008.
233. Montagnese TA, Reader A, Melfi R: A comparative study of the Gow-Gates technique and a standard technique for mandibular anesthesia, *J Endod* 10:158, 1984.
234. Moore KD, Reader A, Meyers WJ, et al: A comparison of the periodontal ligament injection using 2% lidocaine with 1:100,000 epinephrine and saline in human mandibular premolars, *Anesth Prog* 34:181, 1987.
235. Moore P: Long-acting local anesthetics: a review of clinical efficacy in dentistry, *Compendium* 11:24, 1990.
236. Moore P, Werther JR, Seldin EB, et al: Analgesic regimens for third molar surgery: pharmacologic and behavioral considerations, *J Am Dent Assoc* 113:739, 1986.
237. Moore PA, Dunsky JL: Bupivacaine anesthesia: a clinical trial for endodontic therapy, *Oral Surg Oral Med Oral Pathol* 55:176, 1983.
238. Moore PA, Hersh EV: Combining ibuprofen and acetaminophen for acute pain management after third-molar extractions: Translating clinical research to dental practice, *J Am Dent Assoc* 144:898, 2013.
239. Moore TJ, Walsh CS, Cohen MR: Reported adverse event cases of methemoglobinemia associated with benzocaine products, *Arch Intern Med* 164:1192, 2004.
240. Morais-Almeida M, Gaspar A, Marinho S, et al: Allergy to local anesthetics of the amide group with tolerance to procaine, *Allergy* 58:827, 2003.
241. Morse D, Furst ML, Belott RM, et al: Infectious flare-ups and serious sequelae following endodontic treatment: a prospective randomized trial on efficacy of antibiotic prophylaxis in cases of asymptomatic pulpal-periapical lesions, *Oral Surg Oral Med Oral Pathol* 64:96, 1987.
242. Morse D, Furst ML, Belott RM, et al: Prophylactic penicillin versus erythromycin taken at the first sign of swelling in cases of asymptomatic pulpal-periapical lesions: a comparative analysis, *Oral Surg Oral Med Oral Pathol* 65:228, 1988.
243. Moskow A, Morse DR, Krasner P, et al: Intracanal use of a corticosteroid solution as an endodontic anodyne, *Oral Surg Oral Med Oral Pathol* 58:600, 1984.
244. Naftalin L, Yagiela J: Vasoconstrictors: indications and precautions, *Dent Clin North Am* 46:733, 2002.
245. Niv D: Intraoperative treatment of postoperative pain. In Campbell JN, editor: *Pain 1996: an updated review*, Seattle, 1996, IASP Press.
246. Nobuhara WK, Carnes DL, Gilles JA: Anti-inflammatory effects of dexamethasone on periapical tissues following endodontic overinstrumentation, *J Endod* 19:501, 1993.
247. Nusstein J, Berlin J, Reader A, et al: Comparison of injection pain, heart rate increase and post-injection pain of articaine and lidocaine in a primary intraligamentary injection administered with a computer-controlled local anesthetic delivery system, *Anesth Prog* 51:126, 2004.
248. Nusstein J, Burns Y, Reader A, et al: Injection pain and postinjection pain of the palatal-anterior superior alveolar injection, administered with the Wand Plus system, comparing 2% lidocaine with 1:100,000 epinephrine to 3% mepivacaine, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 97:164, 2004.
249. Nusstein J, Claffey E, Reader A, et al: Anesthetic effectiveness of the supplemental intraligamentary injection, administered with a computer-controlled local anesthetic delivery system, in patients with irreversible pulpitis, *J Endod* 31:354, 2005.
250. Nusstein J, Kennedy S, Reader A, et al: Anesthetic efficacy of the supplemental X-tip intraosseous injection in patients with irreversible pulpitis, *J Endod* 29:724, 2003.
251. Nusstein J, Lee S, Reader A, et al: Injection pain and postinjection pain of the anterior middle superior alveolar injection administered with the Wand or conventional syringe, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 98:124, 2004.
252. Nusstein J, Reader A, Beck FM: Anesthetic efficacy of different volumes of lidocaine with epinephrine for inferior alveolar nerve blocks, *Gen Dent* 50:372; quiz, 376; 2002.
253. Nusstein J, Reader A, Nist R, et al: Anesthetic efficacy of the supplemental intraosseous injection of 2% lidocaine with 1:100,000 epinephrine in irreversible pulpitis, *J Endod* 24:487, 1998.
254. Nusstein J, Wood M, Reader A, et al: Comparison of the degree of pulpal anesthesia achieved with the intraosseous injection and infiltration injection using 2% lidocaine with 1:100,000 epinephrine, *Gen Dent* 53:50, 2005.
255. Nusstein JM, Beck M: Effectiveness of 20% benzocaine as a topical anesthetic for intraoral injections, *Anesth Prog* 50:159, 2003.
256. Nuzum FM, Drum M, Nusstein J, et al: Anesthetic efficacy of articaine for combination labial plus lingual infiltrations versus labial infiltration in the mandibular lateral incisor, *J Endod* 36:952, 2010.
257. Oleson M, Drum M, Reader A, et al: Effect of preoperative ibuprofen on the success of the inferior alveolar nerve block in patients with irreversible pulpitis, *J Endod* 36:379, 2010.
258. Pacira Pharmaceuticals, Inc. *How DepoFoam works.* http://www.pacira.com/depofoam-platform/how-it-works.php. Accessed June 23, 2015
259. Papagiannopoulou P, Argiriadou H, Georgiou M, et al: Preincisional local infiltration of levobupivacaine vs ropivacaine for pain control after laparoscopic cholecystectomy, *Surg Endosc* 17:1961, 2003.
260. Parente SA, Anderson RW, Herman WW, et al: Anesthetic efficacy of the supplemental intraosseous injection for teeth with irreversible pulpitis, *J Endod* 24:826, 1998.
261. Parirokh M, Sadeghi A, Nakhaee N, et al: Effect of topical anesthesia on pain during infiltration injection and success of anesthesia for maxillary central incisors, *J Endod* 38:1553, 2012.
262. Parirokh M, Yosefi MH, Nakhaee N, et al: The success rate of bupivacaine and lidocaine as anesthetic agents in inferior alveolar nerve block in teeth with irreversible pulpitis without spontaneous pain, *Restor Dent Endod* 40:155, 2015.
263. Penniston S, Hargreaves K: Evaluation of periapical injection of ketorolac for management of endodontic pain, *J Endod* 22:55, 1996.
264. Pereira LA, Groppo FC, Bergamaschi CD, et al: Articaine (4%) with epinephrine (1:100,000 or 1:200,000) in intraosseous injections in symptomatic irreversible pulpitis of mandibular molars: anesthetic efficacy and cardiovascular effects, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 116:e85, 2013.
265. Pertot W, Dejou J: Bone and root resorption: effects of the force developed during periodontal ligament injections in dogs, *Oral Surg Oral Med Oral Pathol* 74:357, 1992.
266. Peterson J, Matsson L, Nation W: Cementum and epithelial attachment response to the sulcular and periodontal ligament injection techniques, *Pediatr Dent* 5:257, 1983.
267. Peurach J: Pulpal response to intraligamentary injection in cynomolgus monkey, *Anesth Prog* 32:73, 1985.
268. Pfeil L, Drum M, Reader A, et al: Anesthetic efficacy of 1.8 milliliters and 3.6 milliliters of 2% lidocaine with 1:100,000 epinephrine for posterior superior alveolar nerve blocks, *J Endod* 36:598, 2010.
269. Piccinni C, Gissi DB, Gabusi A, et al: Paraesthesia after local anaesthetics: an analysis of reports to the FDA Adverse Event Reporting System, *Basic Clin Pharmacol Toxicol* 117:52, 2015.
270. Plamondon T, Walton R, Graham G, et al: Pulp response to the combined effects of cavity preparation and periodontal ligament injection, *Oper Dent* 15:86, 1990.
271. Pogrel M: Permanent nerve damage from inferior alveolar nerve blocks: an update to include articaine, *J Calif Dent Assoc* 35:217, 2007.
272. Pogrel M: Permanent nerve damage from inferior alveolar nerve blocks: a current update, *J Calif Dent Assoc* 40:795, 2012.
273. Poorni S, Veniashok B, Senthilkumar AD, et al: Anesthetic efficacy of four percent articaine for pulpal anesthesia by using inferior alveolar nerve block and buccal infiltration techniques in patients with irreversible pulpitis: a prospective randomized double-blind clinical trial, *J Endod* 37:1603, 2011.
274. Primosch RE, Robinson L: Pain elicited during intraoral infiltration with buffered lidocaine, *Am J Dent* 9:5, 1996.

275. Quercia R, Voleman, CI: Liposomal bupivacaine: a long-acting local anesthetic for postsurgical analgesia, *Clin Pharmacol* 47:212, 2012.
276. Rawson R, Orr D: Vascular penetration following intraligamental injection, *J Oral Maxillofac Surg* 43:600, 1985.
277. Reader A, Nusstein J, Drum M: *Successful local anesthesia for restorative dentistry and endodontics*, ed 2. Hanover Park, IL, 2017, Quintessence.
278. Reemers T, Glickman G, Spears R, et al: The efficacy of the IntraFlow intraosseous injection as a primary anesthesia technique, *J Endod* 34:280, 2008.
279. Reisman D, Reader A, Nist R, et al: Anesthetic efficacy of the supplemental intraosseous injection of 3% mepivacaine in irreversible pulpitis, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 84:676, 1997.
280. Reitz J, Reader A, Nist R, et al: Anesthetic efficacy of a repeated intraosseous injection given 30 minutes following an inferior alveolar nerve block/intraosseous injection, *Anesth Prog* 45:143, 1998.
281. Reitz J, Reader A, Nist R, et al: Anesthetic efficacy of the intraosseous injection of 0.9 mL of 2% lidocaine (1:100,000 epinephrine) to augment an inferior alveolar nerve block, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 86:516, 1998.
282. Replogle K, Reader A, Nist R, et al: Anesthetic efficacy of the intraosseous injection of 2% lidocaine (1:100,000 epinephrine) and 3% mepivacaine in mandibular first molars, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 83:30, 1997.
283. Replogle K, Reader A, Nist R, et al: Cardiovascular effects of intraosseous injections of 2% lidocaine with 1:100,000 epinephrine and 3% mepivacaine, *J Am Dent Assoc* 130:649, 1999.
284. Roahen JO, Marshall FJ: The effects of periodontal ligament injection on pulpal and periodontal tissues, *J Endod* 16:28, 1990.
285. Robertson D, Nusstein J, Reader A, et al: Anesthetic efficacy of articaine and lidocaine in buccal infiltration injections of the mandibular first molar, *J Am Dent Assoc* 138:1104, 2007.
286. Robison SF, Mayhew RB, Cowan RD, et al: Comparative study of deflection characteristics and fragility of 25-, 27-, and 30-gauge short dental needles, *J Am Dent Assoc* 109:920, 1984.
287. Rogers MJ, Johnson BR, Remeikis NA, et al: Comparison of effect of intracanal use of ketorolac tromethamine and dexamethasone with oral ibuprofen on post treatment endodontic pain, *J Endod* 25:381, 1999.
288. Rollason V, Samer C, Piguet V, et al: Pharmacogenetics of analgesics: toward the individualization of prescription, *Pharmacogenomics* 9:905, 2008.
289. Rood J: The nerve supply of the mandibular incisor region, *Br Dent J* 143:227, 1977.
290. Rood JP: Adverse reaction to dental local anesthetic injection: "allergy" is not the cause, *Br Dent J* 189:380, 2000.
291. Rosenberg P: Clinical strategies for managing endodontic pain, *Endod Topics* 3:78, 2002.
292. Rosenquist J, Rosenquist K, Lee P: Comparison between lidocaine and bupivacaine as local anesthetics with diflunisal for postoperative pain control after lower third molar surgery, *Anesth Prog* 35:1, 1988.
293. Roy M, Nakanishi T: Differential properties of tetrodotoxin-sensitive and tetrodotoxin-resistant sodium channels in rat dorsal root ganglion neurons, *J Neurosci* 12:2104, 1992.
294. Ryan JF, Jureidini B, Hodges JS, et al: Gender differences in analgesia for endodontic pain, *J Endod* 34:552, 2008.
295. Saadoun AP, Malamed S: Intraseptal anesthesia in periodontal surgery, *J Am Dent Assoc* 111:249, 1985.
296. Saatchi M, Khademi A, Baghaei B, et al: Effect of sodium bicarbonate-buffered lidocaine on the success of inferior alveolar nerve block for teeth with symptomatic irreversible pulpitis: a prospective, randomized double-blind study, *J Endod* 41:33, 2015.
297. Saatchi M, Shafiee M, Khademi A, et al: Anesthetic efficacy of Gow-Gates nerve block, inferior alveolar nerve block, and their combination in mandibular molars with symptomatic irreversible pulpitis: a prospective, randomized clinical trial, *J Endod* 44:384, 2018.
298. Salomen M, Forsell H, Sceinin M: Local dental anesthesia with lidocaine and adrenalin: effects on plasma catecholamines, heart rate, and blood pressure, *Int J Oral Maxillofac Surg* 17:392, 1988.
299. Sampaoi RM, Carnaval TG, Lanfredi CB, et al: Comparison of the anesthetic efficacy between bupivacaine and lidocaine in patients with irreversible pulpitis of mandibular molar, *J Endod* 38:594, 2012.
300. Schleder JR, Reader A, Beck M, et al: The periodontal ligament injection: a comparison of 2% lidocaine, 3% mepivacaine, and ‚Äâ1:100,000 epinephrine to 2% lidocaine with 1:100,000 epinephrine in human mandibular premolars, *J Endod* 14:397, 1988.
301. Schwab JM, Schluesener HJ, Meyermann R, et al: COX-3 the enzyme and the concept: steps towards highly specialized pathways and precision therapeutics? *Prostaglandins Leukot Essent Fatty Acids* 69:339, 2003.
302. Scott J, Drum M, Reader A, et al: Efficacy of a repeated infiltration to prolong duration of pulpal anesthesia in maxillary lateral incisors, *J Am Dent Assoc* 140:318, 2009.
303. Seng G, Kraus K, Cartridge G: Confirmed allergic reactions to amide local anesthetics, *Gen Den* 44:52, 1996.
304. Shahi S, Rahimi S, Yavari HR, et al: Success rate of 3 injection methods with articaine for mandibular first molars with symptomatic irreversible pulpitis: a CONSORT randomized double-blind clinical trial, *J Endod* 44:1462, 2018.
305. Shapiro MR, McDonald NJ, Gardner RJ, et al: Efficacy of articaine versus lidocaine in supplemental infiltration for mandibular first versus second molars with irreversible pulpitis: a prospective, randomized, double-blind clinical trial, *J Endod* 44:523, 2018.
306. Sherman MG, Flax M, Namerow K, et al: Anesthetic efficacy of the Gow-Gates injection and maxillary infiltration with articaine and lidocaine for irreversible pulpitis, *J Endod* 34:656, 2008.
307. Shojaei A, Haas D: Local anesthetic cartridges and latex allergy: a literature review, *J Can Dent Assoc* 68:622, 2002.
308. Shoji N, Enod Y, Iikubo M, et al: Dentin hypersensitivity-like tooth pain seen in patients receiving steroid therapy: An exploratory study, *Pharm Sci* 2016: 132:187, 2016.
309. Simon F, Reader A, Drum M, et al: A prospective, randomized single-blind study of the anesthetic efficacy of the inferior alveolar nerve block administered with a peripheral nerve stimulator, *J Endod* 36:429, 2010.
310. Simpson M, Drum M, Reader A, et al: Effect of preoperative ibuprofen/acetaminophen on the success of the inferior alveolar nerve block in patients with symptomatic irreversible pulpitis, *J Endod* 37:593, 2011.
311. Singla M, Subbiya A, Aggarwal V, et al: Comparison of the anesthetic efficacy of different volumes of 4% articaine (1.8 and 3.6 mL) as supplemental buccal infiltration after failed inferior alveolar nerve block, *Int Endod J* 48:103, 2015.
312. Sinnott CJ, Strichartz GR: Levobupivacaine versus ropivacaine for sciatic nerve block in the rat, *Reg Anesth Pain Med* 28:294, 2003.
313. Siqueira J, Barnett F: Interappointment pain: mechanisms, diagnosis, and treatment, *Endod Topics* 3:93, 2004.
314. Sisk A: Evaluation of the Akinosi mandibular block technique in oral surgery, *Oral Maxillofac Surg* 44:113, 1986.
315. Smith G, Pashley D: Periodontal ligament injection: evaluation of systemic effects, *Oral Surg Oral Med Oral Pathol* 56:571, 1983.
316. Smith G, Walton R: Periodontal ligament injections: distribution of injected solutions, *Oral Surg Oral Med Oral Pathol* 55:232, 1983.
317. Smith G, Walton R, Abbott B: Clinical evaluation of periodontal ligament anesthesia using a pressure syringe, *J Am Dent Assoc* 107:953, 1983.
318. Smolinske SC: Review of parenteral sulfite reactions, *J Toxicol Clin Toxicol* 30:597, 1992.
319. Sood R, Hans MK, Shetty S: Comparison of anesthetic efficacy of 4% articaine with 1:100,000 epinephrine and 2% lidocaine with 1:80,000 epinephrine for inferior alveolar nerve block in patients with irreversible pulpitis, *J Clin Exp Dent* 6:520, 2014.
320. Sorensen H, Skidmore L, Rzasa R, et al: Comparison of pulpal sodium channel density in normal teeth to diseased teeth with severe spontaneous pain (abstract), *J Endod* 30:287, 2004.
321. Spiegel BM, Chiou CF, Ofman JJ: Minimizing complications from nonsteroidal antiinflammatory drugs: cost-effectiveness of competing strategies in varying risk groups, *Arthritis Rheum* 53:185, 2005.
322. Stabile P, Reader A, Gallatin E, et al: Anesthetic efficacy and heart rate effects of the intraosseous injection of 1.5% etidocaine (1:200,000 epinephrine) after an inferior alveolar nerve block, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 89:407, 2000.
323. Stambaugh J, Drew J: The combination of ibuprofen and oxycodone/acetaminophen in the management of chronic cancer pain, *Clin Pharmacol Ther* 44:665, 1988.

324. Steinkruger G, Nusstein J, Reader A, et al: The significance of needle bevel orientation in success of the inferior alveolar nerve block, *J Am Dent Assoc* 137:1685, 2006.
325. Strichartz G: Molecular mechanisms of nerve block by local anesthetics, *Anesthesiology* 45:421, 1967.
326. Svensson CI, Yaksh TL: The spinal phospholipase-cyclooxygenase-prostanoid cascade in nociceptive processing, *Annu Rev Pharmacol Toxicol* 42:553, 2002.
327. Taggar TJ, Wu D, Khan AA. A randomized clinical trial comparing two ibuprofen formulations in patients with acute odontogenic pain, *J Endod* 43:674, 2017.
328. Teplitsky P, Hablichek C, Kushneriuk J: A comparison of bupivacaine to lidocaine with respect to duration in the maxilla and mandible, *J Can Dent Assoc* 53:475, 1987.
329. Todorovic L, Stajcic Z, Petrovic V: Mandibular versus inferior alveolar dental anaesthesia: clinical assessment of 3 different techniques, *Int J Oral Maxillofac Surg* 15:733, 1986.
330. Tolas AG, Pflug AE, Halter JB: Arterial plasma epinephrine concentrations and hemodynamic responses after dental injection of local anesthetic with epinephrine, *J Am Dent Assoc* 104:41, 1982.
331. Torabinejad M, Dorn SO, Eleazer PD, et al: Effectiveness of various medications on postoperative pain following root canal obturation, *J Endod* 20:427, 1994.
332. Tortamano IP, Siviero M, Costa CG, et al: A comparison of the anesthetic efficacy of articaine and lidocaine in patients with irreversible pulpitis, *J Endod* 35:165, 2009.
333. Trope M: Relationship of intracanal medicaments to endodontic flare-ups, *Endod Dent Traumatol* 6:226, 1990.
334. Troullos E, Freeman R, Dionne R: The scientific basis for analgesic use in dentistry, *Anesth Prog* 33:123, 1986.
335. Troullos ES, Goldstein DS, Hargreaves KM, et al: Plasma epinephrine levels and cardiovascular response to high administered doses of epinephrine contained in local anesthesia, *Anesth Prog* 34:10, 1987.
336. Troullos ES, Hargreaves KM, Goldstein DS, et al: Epinephrine suppresses stress-induced increases in plasma immunoreactive beta-endorphin in humans, *J Clin Endocrinol Metab* 69:546, 1989.
337. Turner CL, Eggleston GW, Lunos S, et al: Sniffing out endodontic pain: use of an intranasal analgesic in a randomized clinical trial, *J Endod* 37:439, 2011.
338. Vanderheyden PJ, Williams RA, Sims TN: Assessment of ST segment depression in patients with cardiac disease after local anesthesia, *J Am Dent Assoc* 119:407, 1989.
339. VanGheluwe J, Walton R: Intrapulpal injection: factors related to effectiveness, *Oral Surg Oral Med Oral Pathol* 19:38, 1997.
340. Vreeland DL, Reader A, Beck M, et al: An evaluation of volumes and concentrations of lidocaine in human inferior alveolar nerve block, *J Endod* 15:6, 1989.
341. Wali M, Drum M, Reader A, Nusstein J: Prospective, randomized, single-blind study of the anesthetic efficacy of 1.8 and 3.6 milliliters of 2% lidocaine with 1:50,000 epinephrine for the inferior alveolar nerve block, *J Endod* 36:1459, 2010.
342. Wallace J: Selective COX-2 inhibitors: Is the water becoming muddy? *Trends Pharmacol Sci* 20:4, 1999.
343. Wallace JA, Michanowicz AE, Mundell RD, et al: A pilot study of the clinical problem of regionally anesthetizing the pulp of an acutely inflamed mandibular molar, *Oral Surg Oral Med Oral Pathol* 59:517, 1985.
344. Walton R, Abbott B: Periodontal ligament injection: a clinical evaluation, *J Am Dent Assoc* 103:571, 1981.
345. Walton R, Chiappinelli J: Prophylactic penicillin: effect on posttreatment symptoms following root canal treatment of asymptomatic periapical pathosis, *J Endod* 19:466, 1993.
346. Walton RE: Distribution of solutions with the periodontal ligament injection: clinical, anatomical, and histological evidence, *J Endod* 12:492, 1986.
347. Warren CA, Mok L, Gordon S, et al: Quantification of neural protein in extirpated tooth pulp, *J Endod* 34:7, 2008.
348. Webster S Jr, Drum M, Reader A, et al: How effective is supplemental intraseptal anesthesia in patients with symptomatic irreversible pulpitis? *J Endod* 42:1453, 2016.
349. Wells JE, Bingham V, Rowland KC, et al: Expression of Nav1.9 channels in human dental pulp and trigeminal ganglion, *J Endod* 33:1172, 2007.
350. Wells LK, Drum M, Nusstein J, et al: Efficacy of Ibuprofen and ibuprofen/acetaminophen on postoperative pain in symptomatic patients with a pulpal diagnosis of necrosis, *J Endod* 37:1608, 2011.
351. Whitcomb M, Drum M, Reader A, et al: A prospective, randomized double-blind study of the anesthetic efficacy of sodium bicarbonate buffered 2% lidocaine with 1:100,000 epinephrine in inferior alveolar nerve blocks, *Anesth Prog* 57:59, 2010.
352. White JJ, Reader A, Beck M, et al: The periodontal ligament injection: a comparison of the efficacy in human maxillary and mandibular teeth, *J Endod* 14:508, 1988.
353. Wideman G, Keffer M, Morris E, et al: Analgesic efficacy of a combination of hydrocodone with ibuprofen in postoperative pain, *Clin Pharmacol Ther* 65:66, 1999.
354. Wilburn-Goo D, Lloyd L: When patients become cyanotic: acquired methemoglobinemia, *J Am Dent Assoc* 130:826, 1999.
355. Willett J, Reader A, Drum M, et al: The anesthetic efficacy of diphenhydramine and the combination of diphenhydramine/lidocaine for the inferior alveolar nerve block, *J Endod* 34:1446, 2009.
356. Wilson S, Johns P, Fuller P: The inferior alveolar and mylohyoid nerves: an anatomic study and relationship to local anesthesia of the anterior mandibular teeth, *J Am Dent Assoc* 108:350, 1984.
357. Wolf M, Lichtenstein D, Singh G: Gastrointestinal toxicity of nonsteroidal antiinflammatory drugs, *N Engl J Med* 340:1888, 1999.
358. Wood M, Reader A, Nusstein J, et al: Comparison of intraosseous and infiltration injections for venous lidocaine blood concentrations and heart rate changes after injection of 2% lidocaine with 1:100,000 epinephrine, *J Endod* 31:435, 2005.
359. Woodmansey K: Intraseptal anesthesia: a review of a relevant injection technique, *Gen Dent* 53:418, 2005.
360. Woolf C: Evidence for a central component of post-injury pain hypersensitivity, *Nature* 306:686, 1983.
361. Woolf C: Windup and central sensitization are not equivalent, *Pain* 66:105, 1996.
362. Woolf C: Transcriptional and posttranslational plasticity and the generation of inflammatory pain, *Proc Natl Acad Sci U S A* 96:7723, 1999.
363. Wright C, Antal EJ, Gillespie WR, et al: Ibuprofen and acetaminophen kinetics when taken concurrently, *Clin Pharmacol Ther* 34:707, 1983.
364. Wynn R, Meiller T, Crossley H: *Drug information handbook for dentistry*, Hudson, Ohio, 2000, Lexi-Comp.
365. Wynn RL, Bergman SA, Meiller TF: Paresthesia associated with local anesthetics: a perspective on articaine, *Gen Dent* 51:498, 2003.
366. Yared GM, Dagher FB: Evaluation of lidocaine in human inferior alveolar nerve block, *J Endod* 23:575, 1997.
367. Yingling NM, Byrne BE, Hartwell GR: Antibiotic use by members of the American Association of Endodontists in the year 2000: report of a national survey, *J Endod* 28:396, 2002.
368. Yonchak T, Reader A, Beck M, et al: Anesthetic efficacy of infiltrations in mandibular anterior teeth, *Anesth Prog* 48:55, 2001.
369. Yucel E, Hutchison I: A comparative evaluation of the conventional and closed mouth technique for inferior alveolar nerve block, *Aust Dent J* 40:15, 1995.
370. Zarei M, Ghoddusi J, Sharifi E, et al: Comparison of the anesthetic efficacy of and heart rate changes after periodontal ligament or intraosseous X-tip injection in mandibular molars: a randomized controlled clinical trial, *Int Endod J* 45:921, 2012.
371. Zorian EV, Sharagin NV: Comparative evaluation of the topical action of anesthetics on the dental tissues in experimental conditions, *Stomatologiia* 53:1, 1974.